When Shocks Cascade: Time-Varying Propagation in Armed Conflict Networks

Abstract

In protracted civil wars, when does localized violence begin to propagate beyond the original dyad, and which actors become central to that propagation? We argue that conflict systems can undergo qualitative shifts in their propagation structure as the mapping from past violence to future violence tightens around a shrinking set of actors, concentrating systemic leverage in ways that raw activity levels do not reveal. Recovering these transitions requires separating two dimensions of actor importance: dynamic coupling, how tightly an actor's involvement is conditioned by the prior network state, and systemic leverage, how broadly a bilateral escalation involving that actor transmits through the wider system. To do so, we develop a time-varying bilinear autoregressive network model that estimates, period by period, how the prior period's conflict network shapes expected future interactions. The framework yields coupling trajectories that track when actors become structurally embedded in the lag mapping and propagation profiles that trace how a bilateral escalation ripples through the network over subsequent quarters. Applied to armed-group battle networks in Somalia (1997-2018), the model reveals a transition from fragmented, weakly coupled conflict to a system increasingly organized around Al-Shabaab and government forces. Al-Shabaab-linked escalations generate the largest and most temporally consistent downstream changes, not because Al-Shabaab is the most active group, but because its involvement is most consistently embedded in the system's propagation structure. The concentration of coupling and leverage around a small number of actors represents a qualitative change in how violence spreads, with implications for conflict forecasting and intervention design.

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1 Introduction

How do conflict systems reorganize over time, and when do they become most dangerous? In civil wars, a clash between two actors can trigger retaliation, allied entry, and opportunistic violence by third parties (Buhaug and Rød 2006; Cunningham 2013), yet similar clashes sometimes remain contained. Whether violence propagates depends not just on who fights whom, but on the structure of interdependence linking past interactions to future ones. That structure evolves as wars progress through phases of fragmentation, consolidation, and reorganization (Kalyvas 2006; Staniland 2014). In long-running conflicts such as Somalia's, the same system can transition from a fragmented landscape of clan militias and warlords to one organized around a dominant insurgent-counterinsurgent axis, fundamentally changing how localized violence reverberates through the wider conflict environment. The question is therefore not only what a conflict network looks like at any given moment, but when the mapping from past violence to future violence tightens, and which actors sit at the center of that mapping.

Answering this question requires separating two dimensions of actor importance that are often conflated in applied network analysis. The first is dynamic coupling: how strongly an actor's involvement in the current period is predicted by the prior network state under the estimated lag mapping. An actor can be highly active yet weakly coupled if its engagements do not follow stable lag-dependent patterns. The second is systemic leverage: how strongly and how consistently a standardized perturbation involving an actor is transmitted into broader system-wide changes under the fitted lag dynamics, which can be amplifying or dampening depending on the estimated lag operators in a given phase. An actor can be coupled to the lag structure without generating large downstream propagation if the implied spillovers remain localized.

The cross-classification matters. A high-activity but low-coupling actor looks active but idiosyncratic; a high-coupling but low-leverage actor is predictable but locally contained; only an actor that scores high on both is systemically consequential. Many common summaries of “importance,” including centrality measures and static latent structure estimates, do not directly separate these two dimensions in a time-indexed lag mapping. Recovering both dimensions, and showing how they evolve across conflict phases, is the central goal of this paper.[1]

1.1 Theoretical Expectations

We develop three expectations that guide our empirical analysis and that a time-varying framework is well-positioned to evaluate; all three require a framework capable of tracking lag dependence at meaningful temporal resolution. Each expectation is grounded in established findings from the civil war literature but articulates a prediction about the lag dependence structure of conflict networks that has not been tested directly because existing methods lack the necessary temporal resolution.

First, we expect increasing concentration of dynamic coupling. As protracted conflicts mature, organizational capacity and strategic reach allow a subset of groups to become more tightly embedded in the system’s lagged dependence structure, while others remain peripheral (Weinstein 2007; Staniland 2014). Armed organizations that consolidate through recruitment, territorial control, and the development of command structures create durable interaction patterns that are predictable from the prior network state (Cunningham 2013; Kalyvas 2006). Meanwhile, less organized or more opportunistic actors generate violence that is harder to predict from lagged network ac-

[1]Throughout, “influence” refers to predictive structure in the estimated lag mapping, not the causal effect of intervening on an actor.

tivity (Metternich and Wucherpfennig 2020). We evaluate this by tracking whether coupling becomes more concentrated over time, using concentration summaries of actor-level coupling norms such as the coupling-norm Gini reported in the appendix and the actor trajectories reported in the main text. If coupling remains uniformly distributed across actors throughout the observation period, it would suggest that protracted conflict does not produce the kind of structural consolidation that the organizational literature predicts.

Second, we expect stronger lagged interdependence in conflict than in co-belligerence. Hostile interactions create durable strategic commitments: reputational stakes, revenge cycles, and entrenched territorial disputes that sustain patterns of mutual response across periods (Buhaug and Rød 2006; Walter 2004; Fearon 2004). The underlying logic is that fighting generates information and commitment problems that lock actors into ongoing strategic interdependence (Fearon 1995; Walter 2009). Co-belligerence arrangements, by contrast, tend to be instrumental and fluid, shifting as the balance of power changes and dissolving once the shared threat subsides (Christia 2012). In a time-varying autoregressive framework, these differences should be legible in the estimated lag mapping. Relative to conflict networks, ACLED-coded co-belligerence networks should exhibit weaker lag dependence, reflected in influence weights that are smaller in magnitude and less often credibly distinguishable from zero over time. The innovation variance τ_A^2 provides a complementary within-model diagnostic of how rapidly the lag mapping reorganizes. A static model could detect the level difference in lag dependence between conflict and co-belligerence, but only the time-varying specification reveals whether that difference persists throughout the observation period or is concentrated in particular phases, and whether the conflict lag mapping itself reorganizes more rapidly than its co-belligerence counterpart. The core prediction is about weaker lag dependence in co-belligerence than in conflict for the observed joint-participation measure.

Third, we expect cascade heterogeneity among active actors. High activity does not imply systemic leverage: two actors can be equally active while differing sharply in whether their involvement predicts delayed, system-wide spillovers. Research on conflict networks emphasizes that some actors occupy structural positions that amplify localized violence into broader patterns (Metternich et al. 2013; Braithwaite 2010), while others' involvement remains contained (Dorff, Gallop and Minhas 2020). This distinction has practical stakes: conflict forecasting and early warning systems that weight all active actors equally will overweight locally confined violence and underweight the escalations most likely to cascade. Our propagation profiles distinguish these types by measuring not just the magnitude but the temporal consistency of model-implied downstream propagation in absolute terms, allowing cascades to be amplifying or dampening depending on the estimated lag dynamics in a given phase.

1.2 Existing Approaches and Our Framework

Existing approaches to dynamic networks generally fall into two camps. The first models tie formation and dissolution directly, including relational event models (Butts 2008), temporal exponential random graph models (Hanneke and Xing 2007; Cranmer and Desmarais 2011), and stochastic actor-oriented models (Snijders 1996). These frameworks address micro-processes of tie change but do not directly estimate a time-varying mapping from the lagged network to expected future interactions, which is the object needed to distinguish high-cascade from low-cascade phases.

The second camp models evolving latent structure, but these approaches do not directly estimate a time-indexed mapping from lagged interactions to expected future interactions. Additive and Multiplicative Effects (AME) models (Hoff 2005; Minhas, Hoff and Ward 2019; Hoff 2021) provide latent factor decompositions but assume fixed dependence parameters. Multilinear Tensor Regression approaches (Kolda and Bader 2009; Marrs et al. 2020) generalize to multi-layer networks but treat influence matrices as time-invariant. Recent work relaxes these restrictions: Sewell and Chen (2015) allow latent positions to evolve via Gaussian processes; Durante, Dunson and Vogelstein (2017) learn similarity structures across replicated networks; Kim et al. (2023) extend AME with autoregressive latent positions; Gallop and Minhas (2021) infer time-varying ideal points from diplomatic behavior; and Olivella, Pratt and Imai (2022) propose a dynamic blockmodel with group-based temporal structure. These approaches recover time-varying similarity or position, but encode dependence through contemporaneous latent structure or fixed coefficients rather than a time-indexed bilinear lag operator (Hoff 2015; Minhas, Hoff and Ward 2016; Cranmer et al. 2017). Without a framework that yields both actor-level and system-level diagnostics of evolving lagged dependence, claims about evolving conflict propagation, shifting alliance structures, and the consolidation of armed groups remain difficult to evaluate with the precision that quantitative network data should permit.

In conflict settings, this gap matters because the core substantive claims of the literature, from retaliation cycles and coalition activation to contagion and conflict diffusion, are about how past interactions reorganize future interactions, not only about where actors sit in a similarity space at a given moment (Buhaug et al. 2011). We address this by extending the bilinear autoregressive model (Hoff 2015) so that the sender and receiver influence matrices evolve over time via stochastic processes. Conceptually, the extension is analogous to time-varying parameter VAR models (Primiceri 2005), except that the evolving "coefficients" are structured bilinear operators acting on a lagged network rather than a vector of series. The model yields three diagnostics unavailable from static approaches. Influence weight snapshots $W_t = A_t B_t^\top$ reveal which actor pairs have aligned or opposed structural roles at each period. Coupling norms $\|A_{t,i\cdot}\|_2$ track how tightly each actor's next-period involvement is predicted by the lagged network. Propagation profiles measure the model-implied downstream change in system-wide intensity following a standardized bilateral escalation, capturing systemic leverage. Together, these answer related but distinct questions: who is aligned with whom, whose involvement is most tightly conditioned by lagged dynamics, and whose escalations are associated with the broadest predicted downstream consequences.

We apply the model to quarterly armed-group battle networks in Somalia from 1997-2018, constructed from the Armed Conflict Location and Event Dataset (ACLED). Somalia's protracted conflict offers a compelling setting because it spans several distinct phases: clan-based fragmentation and warlord competition in the late 1990s, the brief consolidation under the Islamic Courts Union in 2006, Al-Shabaab's emergence as the dominant insurgent organization after 2007, and the gradual expansion of AMISOM and government authority thereafter (Menkhaus 2007; Marchal 2009; Hansen 2013; Williams 2018). These transitions should produce precisely the kind of time-varying propagation structure our model is designed to detect. The results are consistent with each of our theoretical expectations. The estimated lag mapping shifts from diffuse and multipolar to increasingly concentrated around Al-Shabaab and government forces. Conflict networks exhibit substantially stronger and more persistent lagged interdependence than co-belligerence networks. And among all actors, Al-Shabaab-linked escalations generate the largest and most temporally stable predicted downstream effects; this is not because Al-Shabaab is simply the most active group, but because its involvement is most consistently embedded in a propagation structure that maps bilateral escalation into multi-quarter, network-wide spillovers.

This paper makes three contributions. First, substantively, it demonstrates that Somalia's conflict propagation structure reorganized around a shrinking set of actors over two decades, and that only Al-Shabaab developed the combination of sustained coupling and reliable cascade generation that marks a systemically consequential actor. This finding advances our understanding of how protracted civil wars evolve: the shift toward concentrated propagation represents a qualitative change in how violence spreads through the conflict system, with implications for conflict forecasting and intervention design. Second, methodologically, it introduces a dynamic bilinear autoregressive network model with fully conjugate Bayesian inference, extending the static bilinear framework of Hoff (2015) to time-varying influence matrices while preserving computational tractability. The approach applies to any political network where lagged interdependence is expected to vary over time. Third, analytically, it decomposes actor importance into coupling and leverage, showing that these dimensions are conceptually distinct and empirically separable across conflict phases.

2 Static Bilinear Autoregressive Model

We begin by reviewing the static bilinear autoregressive model (Hoff 2015; Minhas, Hoff and Ward 2016), which provides the foundation for our dynamic extension. It estimates fixed sender and receiver influence matrices that map the previous period's network into expected future interactions, establishing the core bilinear structure before we allow that structure to evolve over time. For $t = 1, \ldots, T$, let $Y_t \in \mathbb{R}^{n \times n}$ record dyadic interaction intensities (in the application below, quarterly event counts between actor pairs, treated as scaled intensities for modeling purposes) and $\Theta_t \in \mathbb{R}^{n \times n}$ represent the de-noised latent interaction state that evolves according to lagged network dependence.[2] For $t = 2, \ldots, T$, the latent process evolves according to the bilinear autore-

[2]The Gaussian observation model treats event counts as continuous intensities. This is an approximation; our substantive diagnostics depend on the cross-dyadic propagation structure recovered by the

gression:

$$\Theta_t = A\Theta_{t-1}B^\top + \sigma E_t, \quad E_t \overset{\text{i.i.d.}}{\sim} \text{MN}_{n,n}(0, I_n, I_n), \tag{1}$$

where $A, B \in \mathbb{R}^{n\times n}$ are time-invariant influence matrices, σ^2 is the process variance, and MN denotes a matrix-normal distribution. This static specification restricts the influence matrices A and B to remain constant across all time periods, averaging influence effects over the entire observation window. The bilinear decomposition separates two influence mechanisms through the element-wise representation:

$$\Theta_{i,j,t} = \sum_{i'}\sum_{j'} a_{i,i'} b_{j,j'} \Theta_{i',j',t-1} + \sigma\epsilon_{i,j,t}.$$

In a conflict setting, A captures how past sending behavior influences future sending patterns: a large $a_{i,i'}$ indicates that group i's current involvement is shaped by the prior actions of group i', whether through emulation, retaliation, or strategic response. The matrix B captures how prior targeting behavior affects future targets: a high $b_{j,j'}$ implies that actors previously targeting j' are now more likely to target j, reflecting shifting alliances or retaliation dynamics. Together, A and B decompose the temporal evolution of the conflict network into interpretable influence patterns among aggressors and among targets.[3]

The observed data relate to the latent intensities through an additive observation equation:

lag operator, not on the distributional form of individual cell counts.

[3]In undirected settings the sender-receiver distinction is not substantively meaningful, but we retain A and B (and, in the dynamic model, A_t and B_t) to preserve a conjugate updating scheme. In practice they recover similar structure, and our substantive diagnostics are invariant to how the bilinear product is partitioned between them.

$$Y_t = M + \Theta_t + V_t, \quad V_t \overset{\text{i.i.d.}}{\sim} \mathsf{MN}_{n,n}(0, \sigma_Y^2 I_n, I_n), \tag{2}$$

where $M \in \mathbb{R}^{n \times n}$ is an unrestricted baseline of dyad-specific intercepts, updated element-wise, that absorbs persistent rivalry, persistent coordination, and stable reporting differences across dyads, so that the influence structure in A and B reflects dynamic propagation rather than static heterogeneity. [4]

The static model provides an interpretable decomposition of lagged network dependence, but it imposes a single influence structure across the entire observation window. In a conflict spanning two decades and multiple strategic regimes, a time-averaged influence matrix cannot distinguish periods of fragmented, multipolar competition from periods of concentrated, bipolar confrontation. When the underlying dynamics shift, as they do across phases of a protracted conflict, this averaging obscures precisely the variation of interest: when influence concentrates, when cascades become more likely, and which actors drive those transitions.

3 Time-Varying Bilinear Autoregressive Model

In civil wars, who echoes whom changes as alliances form and dissolve, organizations consolidate or fragment, and external interventions reshape the strategic environment. A peacekeeping deployment, a successful offensive, or a regime change can rewire the lagged dependence structure from one period to the next. Fixing the influence matrices over a long horizon averages away these shifts, which are often the primary object of substantive interest.

[4] The observation variance σ_Y^2 may be fixed to 1 for identifiability. For the initial state we adopt a diffuse prior $\Theta_1 \sim \mathsf{MN}_{n,n}(0, \kappa_\Theta^2 I_n, I_n)$ with $\kappa_\Theta^2 \gg 1$. Conjugate priors on all remaining parameters yield a Gibbs sampler with closed-form full conditionals; derivations appear in the appendix.

Our extension to Hoff (2015) and Minhas, Hoff and Ward (2016) allows the sender and receiver influence matrices to evolve over time. This yields time-indexed diagnostics, including influence snapshots, actor-level coupling trajectories, and period-specific propagation profiles, while the inference strategy remains tractable within a conjugate Bayesian state-space framework through row-wise Forward-Filtering Backward-Sampling (FFBS; Carter and Kohn 1994). Let $A_t, B_t \in \mathbb{R}^{n\times n}$ represent time-varying sender and receiver influence matrices. The latent network dynamics become:

$$\Theta_t = A_t \Theta_{t-1} B_t^\top + \sigma E_t, \quad E_t \overset{\text{i.i.d.}}{\sim} \mathsf{MN}_{n,n}(0, I_n, I_n), \quad t = 2, \dots, T. \tag{3}$$

The element-wise representation reveals how influence coefficients now evolve:

$$\Theta_{i,j,t} = \sum_{i'} \sum_{j'} a_{i,i',t} b_{j,j',t} \Theta_{i',j',t-1} + \sigma \epsilon_{i,j,t}.$$

The time-varying parameters retain their interpretations: $a_{i,i',t}$ represents the time-specific influence of actor i''s past behavior on actor i's current behavior, while $b_{j,j',t}$ captures the time-specific effect of being targeted. Unlike the static model, these influence relationships can now strengthen, weaken, or reverse over time. To model this temporal evolution while maintaining computational tractability, we impose matrix-element random walks on A_t and B_t:

$$a_{k,\ell,t} = a_{k,\ell,t-1} + \eta^A_{k,\ell,t}, \quad \eta^A_{k,\ell,t} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \tau_A^2), \tag{4}$$

$$b_{k,\ell,t} = b_{k,\ell,t-1} + \eta^B_{k,\ell,t}, \quad \eta^B_{k,\ell,t} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \tau_B^2). \tag{5}$$

Equivalently, each row of A_t evolves as an n-dimensional random walk with innovation covariance $\tau_A^2 I_n$ (and similarly for B_t), which is the representation exploited for row-wise

FFBS in posterior inference.

For initial conditions, we specify diffuse priors: $A_1 \sim \mathsf{MN}_{n,n}(0, \kappa_A^2 I_n, I_n)$ and $B_1 \sim \mathsf{MN}_{n,n}(0, \kappa_B^2 I_n, I_n)$.

The innovation variances τ_A^2 and τ_B^2 control the degree of temporal variation; the random walk serves as a smoothing prior that borrows strength across adjacent periods and shrinks influence estimates toward gradual evolution unless the data strongly support abrupt change. Small values enforce gradual evolution while larger values permit rapid adaptation to structural breaks. Although the prior favors smoothness, the random walk does not preclude sharp shifts when supported by the likelihood; periods of rapid reorganization appear as spikes in posterior innovations $a_{k,\ell,t} - a_{k,\ell,t-1}$. In practice, the posterior estimate of τ_A^2 provides a data-driven measure of how rapidly the conflict system's influence structure reorganizes: a large posterior mean indicates that the data support substantial quarter-to-quarter variation in the lag mapping, while a small value indicates that a nearly static structure suffices. Because A_t and B_t are only identified up to reciprocal rescaling, τ_A^2 and τ_B^2 should be interpreted primarily as within-model diagnostics of temporal smoothness under the adopted prior scaling; the main cross-model comparisons in the paper emphasize scale-invariant diagnostics derived from W_t and from propagation profiles. As an alternative, mean-reverting AR(1) specifications can replace the random walks: $a_{k,\ell,t} = \rho_A a_{k,\ell,t-1} + \eta^A_{k,\ell,t}$ with $|\rho_A| < 1$.

We complete the hierarchical specification with inverse-gamma priors:

$$\sigma^2 \sim \mathsf{IG}(a_\sigma, b_\sigma), \quad \sigma_Y^2 \sim \mathsf{IG}(a_Y, b_Y), \tag{6}$$

$$\tau_A^2 \sim \mathsf{IG}(a_A, b_A), \quad \tau_B^2 \sim \mathsf{IG}(a_B, b_B). \tag{7}$$

By vectorizing $x_t = \mathsf{vec}(\Theta_t) \in \mathbb{R}^{n^2}$ and defining $F_t = B_t \otimes A_t$, the model becomes a linear-Gaussian state-space system:

$$x_t = F_t x_{t-1} + \sigma \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, I_{n^2}), \tag{8}$$

$$y_t = m + x_t + \sigma_Y \eta_t, \quad \eta_t \sim \mathcal{N}(0, I_{n^2}), \tag{9}$$

where $y_t = \text{vec}(Y_t)$ and $m = \text{vec}(M)$. Intuitively, the Kronecker product simply expresses the bilinear mapping $A_t(\cdot)B_t^\top$ as a standard linear transition in vectorized form, enabling exact conditional updates for $\Theta_{1:T}$ via off-the-shelf Kalman filtering and FFBS, conditional on $A_{1:T}$ and $B_{1:T}$.

3.1 Posterior Inference

The dynamic model admits efficient Gibbs sampling where we draw entire trajectories $(\Theta_{1:T}, A_{1:T}, B_{1:T})$ rather than single matrices. Each block retains conjugacy through the state-space structure. Given the time-varying influence matrices, the latent path $\Theta_{1:T}$ follows a linear-Gaussian state-space model. We apply standard FFBS, consisting of a forward Kalman filtering pass that computes $p(\Theta_t \mid Y_{1:t}, A_{1:t}, B_{1:t})$ for each t, followed by backward sampling of $\Theta_{1:T}$ from the implied smoothing distribution. In implementation we exploit the Kronecker identity $F_t v = \text{vec}(A_t V B_t^\top)$, where V reshapes v into an $n \times n$ matrix, to apply the transition operator without forming $n^2 \times n^2$ matrices explicitly.

The time-varying structure enables a key computational simplification: conditional on $\Theta_{1:T}$ and $B_{1:T}$, each row of A_t evolves as an independent dynamic linear model. Applying FFBS directly to the full n^2-dimensional vectorized system would require $O(n^4)$ covariance updates per time step; the row-wise decomposition replaces this with n independent n-dimensional FFBS passes, reducing the per-sweep cost to $O(n^3T)$ and making the sampler practical for the network sizes encountered in political applications. For row k, define:

$$y_t^{(k)} = \Theta_{k\cdot,t}^\top \in \mathbb{R}^n, \tag{10}$$

$$X_t^{(k)} = (\Theta_{t-1} B_t^\top)^\top \in \mathbb{R}^{n \times n}, \tag{11}$$

$$a_t^{(k)} = A_t[k, \cdot]^\top \in \mathbb{R}^n. \tag{12}$$

This yields the observation equation:

$$y_t^{(k)} = X_t^{(k)} a_t^{(k)} + \varepsilon_t^{(k)}, \quad \varepsilon_t^{(k)} \sim \mathcal{N}(0, \sigma^2 I_n), \tag{13}$$

combined with the random walk state evolution:

$$a_t^{(k)} = a_{t-1}^{(k)} + \omega_t^{(k)}, \quad \omega_t^{(k)} \sim \mathcal{N}(0, \tau_A^2 I_n). \tag{14}$$

We apply FFBS to this n-dimensional DLM for each row $k = 1, \ldots, n$.

Matrix $B_{1:T}$ updates analogously, conditioning on $\Theta_{1:T}$ and $A_{1:T}$ and applying FFBS to each row of B_t. The baseline matrix M and variance parameters $\{\sigma^2, \sigma_Y^2, \tau_A^2, \tau_B^2\}$ retain closed-form conjugate updates (normal and inverse-gamma, respectively); full expressions appear in the appendix.[5] A complete Gibbs sweep proceeds as follows:

1. **Latent path**: Sample $\Theta_{1:T}$ via FFBS given $\{A_{1:T}, B_{1:T}, \sigma^2, \sigma_Y^2, M\}$
2. **Influence matrix A**: Sample rows of $A_{1:T}$ via row-wise FFBS
3. **Influence matrix B**: Sample rows of $B_{1:T}$ via row-wise FFBS

[5]Complete full conditional distributions for all parameters of the dynamic model are provided in the online appendix.

4. **Baseline matrix**: Update each m_{ij} independently
5. **Variance parameters**: Sample $\{\sigma^2, \sigma_Y^2, \tau_A^2, \tau_B^2\}$ from inverse-gamma distributions

3.2 Identification and Relationship to the Static Model

The time-varying extension nests the static model as a special case. Setting $\tau_A^2 = \tau_B^2 = 0$ recovers the static model exactly, while small positive values allow gradual evolution and larger values permit rapid adaptation to regime changes. This nesting clarifies interpretation: when the innovation variances are near zero, the model reduces to a nearly static lag mapping, and when they are larger, the data support more rapid evolution. The posterior distribution of τ_A^2 itself serves as an informal diagnostic: posterior mass concentrated near zero favors the static specification, while mass away from zero indicates that the data require time-varying influence.

Bilinear models exhibit sign and scale ambiguities: a global sign flip $(A_t, B_t) \to (-A_t, -B_t)$ and rescaling $(A_t, B_t) \to (cA_t, c^{-1}B_t)$ both leave the bilinear product unchanged. The influence weight matrix $W_t = A_t B_t^\top$, the transition operator $F_t = B_t \otimes A_t$, and the propagation profiles derived from them are all invariant to these transformations; these are our primary diagnostics. Actor-level coupling summaries based on row norms of A_t are not themselves scale-invariant, so we interpret them comparatively over time and validate them against the invariant row norms of W_t. Both B_t-based summaries and the scale-invariant $\|W_{t,i\cdot}\|_2$ identify the same actors as most structurally embedded (Al-Shabaab and government forces) and the same qualitative temporal patterns (rising concentration over time), confirming that our substantive conclusions do not depend on the particular scale partition between A_t and B_t in any given posterior draw. Sign indeterminacy is resolved through prior initialization and the continuity induced by the evolution process.

The time-varying influence matrices summarize how lagged network patterns are reweighted into current interactions under the fitted dynamics. In the undirected application, we emphasize sign- and scale-invariant diagnostics derived from the influence weight matrix $W_t = A_t B_t^\top$ and from propagation profiles, which indicate whether actors have aligned or opposed lag profiles and how perturbations are transmitted through the estimated sequence of lag operators. At the element level, individual coefficients retain directional interpretations: $a_{i,i',t} > 0$ indicates that actor i loads positively on sender i''s lagged pattern at time t, while $b_{j,j',t} < 0$ suggests that past targeting of j' is associated with reduced targeting of j. At the system level, trajectories of $\|F_t\|_F = \|A_t\|_F \cdot \|B_t\|_F$ measure how overall network memory evolves over time, and posterior draws of innovations $a_{k,\ell,t} - a_{k,\ell,t-1}$ pinpoint when specific influence relationships shift.[6]

The model therefore separates periods of stability from rapid structural change and identifies when specific influence relationships shift, for which actors, and by how much.[7]

[6]At individual time points the spectral radius of $F_t = B_t \otimes A_t$ exceeds 1, which would imply divergence if the operator were held fixed. Because the system is non-autonomous (the transition operator changes every period), pointwise spectral radii do not determine long-run behavior; what matters is the sequence of operators $\{F_\tau\}_{\tau=1}^t$, and the observation equation further anchors each Θ_t to the data. Our propagation profiles propagate perturbations through the estimated sequence of operators, so downstream effects can grow over short horizons when individual operators are expansive.

[7]Formal derivations of the propagation profiles appear in the appendix.

4 Empirical Application: Time-Varying Influence in Armed Group Networks

We apply the framework to conflict and cooperation networks among armed groups in Somalia from 1997 through 2018, constructed from the Armed Conflict Location and Event Dataset (ACLED; Raleigh et al. 2010). Somalia offers a compelling case for several reasons, and in some respects provides a demanding test of our framework. The conflict involves a large number of armed groups operating across clan, ideological, and territorial lines, with up to 34 distinct organizations observed across the two relational dimensions. The sheer multiplicity of actors, combined with the overlap between clan allegiances, Islamist ideology, and territorial competition, generates a complex web of interactions that challenges any model's ability to recover structured lag dependence. The absence of a functioning central state for much of this period allowed multiple power centers to emerge and compete, generating precisely the kind of multipolar-to-bipolar transitions that time-varying influence diagnostics should detect.

The sample spans several distinct phases that the conflict literature has documented: clan-based fragmentation and warlord competition in the late 1990s (Menkhaus 2007); the Ethiopian intervention and brief Islamic Courts Union consolidation in 2006; Al-Shabaab's emergence as the dominant insurgent organization after 2007, drawing on a combination of Islamist ideology, clan networks, and territorial control (Marchal 2009; Hansen 2013); and the gradual expansion of African Union (AMISOM) and government authority thereafter (Williams 2018). If our theoretical expectations hold, the early fragmented period should show diffuse coupling with no dominant actors, while the post-2007 consolidation should show coupling concentrating around Al-Shabaab and government forces. We focus our detailed analysis on four groups based on their high levels of participation in conflict events[8]: Somali Government Forces, Al-Shabaab, Ahlu Sunna Waljama (ASWJ), and Habar Gedir. Appendix figures report coupling and propagation summaries for all 25 actors; the qualitative concentration patterns do not depend on the choice of focal groups. Parallel analyses for the Democratic Republic of Congo and Sudan appear in the appendix.

4.1 Data Construction

We construct two quarter-indexed network series from ACLED battle events. In the conflict network, each cell (i, j, t) records the number of battle events in which actors i and j fought each other during quarter t; since ACLED does not reliably encode conflict initiation and the Actor1 field does not necessarily indicate the aggressor (Armed Conflict Location & Event Data Project (ACLED) N.d.), we treat these ties as symmetric. In the co-belligerence network, each cell records the number of events in which actors i and j jointly participated against a shared opponent. We refer to co-belligerence as the "cooperation network" for brevity.[9] We collapse all state security organs into a single government node, reflecting the fact that the Transitional Federal Government, the Somali National Army, and various regional security forces all operated under a broadly unified chain of command for the purposes of the conflict, even if their coordination was imperfect in practice. We exclude external actors (e.g., AMISOM), so the model estimates a domestic propagation structure. We aggregate to quarters because the mechanisms of interest (retaliation, coalition reconfiguration) operate on the order of weeks to months rather than days.

[8]They have the highest total degree (cumulative number of distinct conflict ties across all quarters) in the conflict network.

[9]Results should be interpreted as evidence about lag dependence in observed joint participation, not as a general claim about all forms of inter-group cooperation.

The Somalia conflict network spans 88 quarterly time points (Q1 1997 through Q4 2018, the endpoint of our data extraction) with a fixed set of 25 actors, defined as the union of all groups that appear in at least one hostile engagement during the period; actors that are inactive in a given quarter simply have all-zero rows and columns. The cooperation network spans 62 time points with 34 actors. The actor sets overlap but are not identical: the cooperation network includes additional groups that are observed only in co-belligerence ties (groups that fight alongside others but are never recorded as directly opposing each other), while the conflict network retains only actors observed in at least one hostile engagement. The networks differ in length because the co-belligerence network includes only quarters with at least one observed co-belligerence tie. Under this convention, a one-step lag corresponds to the previous quarter in which any co-belligerence is observed, not necessarily the previous calendar quarter. This has two competing effects: skipping inactive quarters creates irregular lag lengths, which can attenuate detectable dependence; but it also conditions on quarters when cooperation is present, which is a natural way to ask whether observed co-participation exhibits persistent dependence when it occurs. Even under this conditioning, we recover weights concentrated near zero and little temporal structure, consistent with weak path-dependence in observed co-belligerence.[10] ACLED coverage in Sub-Saharan Africa expanded over time (Eck 2012), and multi-actor battles are dyadized (a battle with m versus ℓ actors contributes $m \times \ell$ conflict dyads); the dyad-specific baseline M absorbs time-invariant reporting differences, and influence diagnostics depend on cross-dyadic co-movement rather than secular count changes. The cell entries $Y_{ij,t}$ are raw event

[10] An alternative would be to keep the full quarterly grid and let non-co-belligerence quarters enter as all-zero matrices. We retain the skip-quarter convention to avoid treating structurally absent data as informative zeros; treating those quarters as zeros yields the same qualitative conclusion (see the appendix).

counts (untransformed); diagonal entries $Y_{ii,t}$ are structurally zero and are excluded from the likelihood (equivalently, we fix $\Theta_{ii,t} = 0$ so that diagonal entries do not contribute to the log-likelihood); all summary statistics are computed over off-diagonal elements only. The "one-unit perturbation" in our propagation profiles corresponds to a one-unit increase in latent intensity on this raw-count scale. Because the perturbation is applied to the latent deviation from the dyad baseline M_{ij}, it should be read as a standardized increase in the model's underlying conflict propensity for that dyad, anchored to the scale of observed counts. For reference, the median non-zero quarterly conflict count between actor pairs is 2, so a one-unit perturbation represents roughly a 50% increase at the median active dyad. Descriptive statistics and parallel analyses for the Democratic Republic of Congo and Sudan (25 actors, 72-94 time points each) appear in the appendix.

4.2 Model Specification

We apply the time-varying bilinear model to both conflict and co-belligerence networks separately, allowing us to examine how influence patterns evolve differently across these two relational dimensions. Although ties are symmetric in these networks (ACLED does not reliably encode conflict initiation, so we treat all interactions as undirected), we retain the general directed specification with separate A_t and B_t matrices to preserve the fully conjugate row-wise FFBS updates described above.[11] For each network type, we estimate:

[11]Because ties are symmetric, one could equivalently write the likelihood over the upper triangle only. We estimate the model on the full $n \times n$ matrix for computational convenience under the conjugate matrix-normal formulation. The estimated observation variance $\hat{\sigma}^2_Y$ adjusts accordingly, and our primary diagnostics (W_t, coupling rankings, propagation profiles) are invariant summaries of the fitted lag operator that do not depend on σ^2_Y directly.

$$Y_t = M + \Theta_t + V_t, \tag{15}$$

$$\Theta_t = A_t \Theta_{t-1} B_t^\top + \sigma E_t, \tag{16}$$

where A_t and B_t follow independent element-wise random walks with innovation variances τ_A^2 and τ_B^2. Because sender and receiver roles are not substantively distinct for undirected ties, the two matrices play exchangeable roles; we report influence diagnostics based on A_t throughout, noting that B_t-based summaries yield qualitatively similar patterns. We use a Gaussian observation model, which keeps inference conjugate and focuses attention on the time-varying cross-dyadic dependence structure rather than on the marginal distribution of individual cell counts (Hoff 2015).[12] We use weakly informative inverse-gamma priors on variance parameters (hyperparameters reported in the appendix).

We run the Gibbs sampler for 10,000 post-burn-in iterations with a burn-in period of 5,000 iterations and thin every 10th draw, yielding 1,000 posterior samples for inference. Trace plots confirming adequate mixing and convergence are provided in the appendix.

We estimate a time-varying lagged dependence mapping and summarize it with three diagnostics, each capturing a different aspect of the propagation structure. Influence weights $W_{ij,t} = \sum_\ell a_{i,\ell,t}\, b_{j,\ell,t}$ measure pairwise alignment or opposition in the system's lag dependence (Section 4.3). Actor coupling norms $\|A_{t,i\cdot}\|_2$ summarize how tightly each actor's next-period involvement is conditioned by the lagged network state

[12] Under this specification, $\Theta_{ij,t}$ is a latent score (deviation from the dyad baseline M_{ij}) that can take negative values; we use "intensity" throughout as shorthand for higher versus lower latent conflict propensity, not as a nonnegative rate. The same bilinear logic can be adapted to ordinal likelihoods (see the appendix for an ordinal extension).

(Section 4.4). Period-specific propagation profiles quantify the model-implied change in conditional mean off-diagonal intensity following a standardized one-unit perturbation, capturing systemic leverage (Section 4.5).

4.3 Evolving Conflict Structure

Figures 1 and 3 display the influence weight matrix $W_t = A_t B_t^\top$ at selected quarters. Positive entries (blue) indicate aligned influence profiles: lagged network patterns that raise one actor's conflict involvement tend also to raise the other's. For instance, a positive weight between Al-Shabaab and a clan militia would indicate that the two actors load on similar lagged dimensions of the fitted mapping, so that the same prior-period network patterns that predict higher involvement for one also predict higher involvement for the other. Negative entries (red) indicate opposed profiles, which can reflect strategic substitution, competitive displacement, or broader reallocation of conflict attention across actors under the fitted lag mapping. These weights should not be interpreted as a direct dyadic effect of actor i on actor j; rather, $W_{ij,t}$ summarizes the extent to which actors i and j load on the same lagged dimensions of the fitted mapping from Θ_{t-1} to Θ_t, so it is a profile-alignment diagnostic. Gray cells represent pairs whose 95% credible intervals contain zero. Because entries of W_t can be large in absolute value, the heat maps truncate the color scale to $[-0.03, 0.03]$ to make the sign pattern visible; untruncated weights often exceed this range, and all summary statistics use untruncated values.

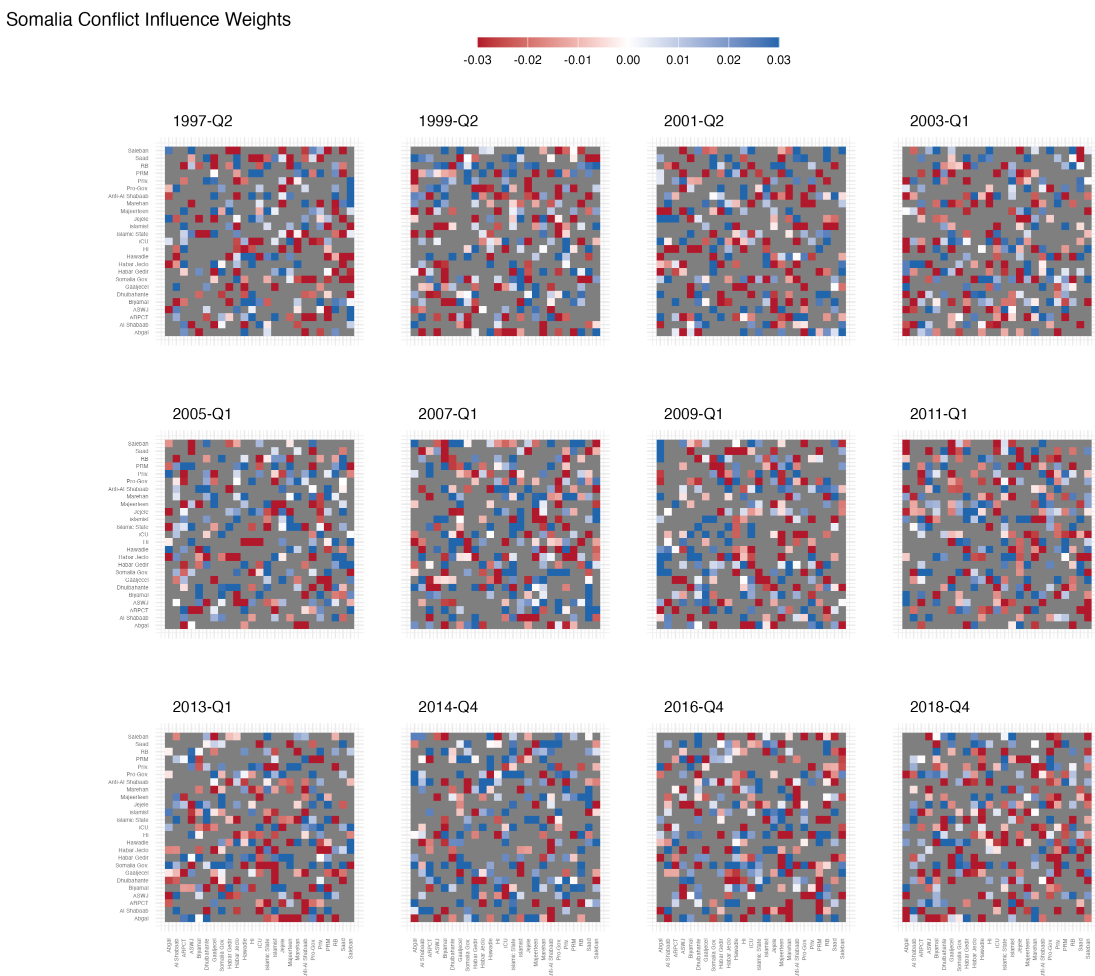

Figure 1: Conflict influence weights $W_t = A_t B_t^\top$ at selected quarters. Color scale truncated to $[-0.03, 0.03]$ for legibility (untruncated weights often exceed this range). Gray cells mark weights with 95% credible intervals containing zero.

The heat maps in Figure 1 reveal that the estimated lag mapping is sparse and unevenly identified: the predominance of gray cells indicates that, for most pairs at most times, the posterior does not sharply distinguish the sign of $W_{ij,t}$ from zero. In a 25-actor conflict system, only a minority of dyadic relationships shows alignment or opposition strong enough to be distinguished from zero given the available data; many dyads are weakly identified and/or weakly coupled in the lag mapping, and the gray cells reflect both possibilities. The substantive pattern is that the subset of weights credibly distinguishable from zero becomes increasingly organized around a small set of actors as the conflict matures, with the same actors recurring across periods in both aligned and opposed relationships. Read together with the coupling trajectories below, this pattern is consistent with a shift toward a propagation regime in which Al-Shabaab and government forces occupy the most consistently consequential structural roles.

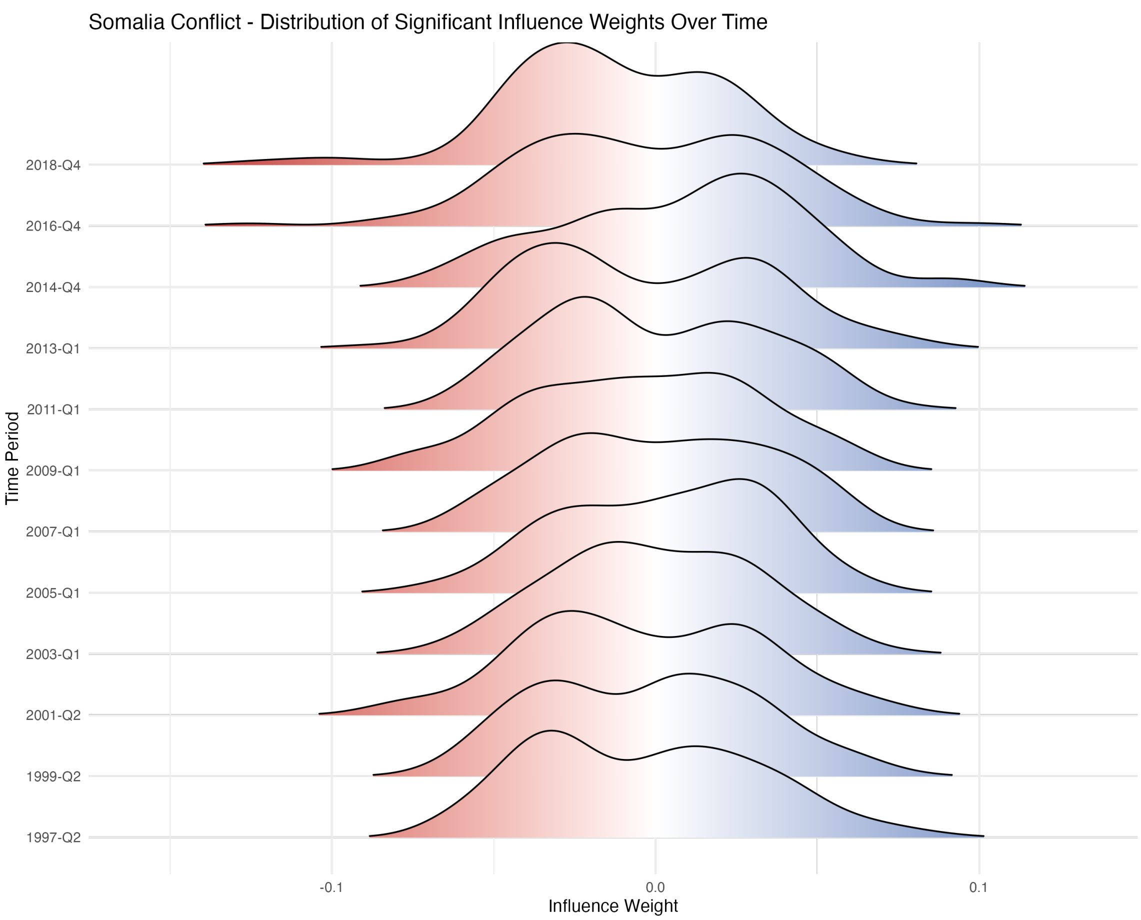

Figure 2: Conflict: evolution of the distribution of credibly different from zero influence weights over time.

Figure 2 provides distributional evidence for this reorganization by plotting the distribution of influence weights whose 95% credible intervals exclude zero. Early in the sample, the subset of credibly nonzero weights is concentrated at small magnitudes. Over time, that subset becomes more polarized, with both stronger positive alignment and stronger negative opposition emerging among a small and increasingly recurring set of actor pairs. A complementary scalar dispersion index computed over all off-diagonal $W_{ij,t}$ values at each time point (including the many weights that remain near zero; see the appendix) shows a clear level difference between conflict and cooperation but does not exhibit a strong monotonic trend within conflict. Read together, these patterns suggest that what changes most is which actor pairs carry the strongest and most clearly identified weights, rather than a uniform, system-wide inflation of weight magnitudes across all dyads. From a forecasting perspective, this reorganization implies that later periods in the conflict are more legible in terms of which pathways dominate the fitted mapping, even as it remains diffuse for most actor pairs.[13]

The timing of this reorganization corresponds to documented shifts in Somalia's conflict environment. The Islamic Courts Union's brief seizure of Mogadishu in 2006 and the Ethiopian military intervention that followed marked a turning point: the fragmented landscape of competing clan militias gave way to a more ideologically organized insurgency as Al-Shabaab consolidated control over much of south-central Somalia (Menkhaus 2007; Marchal 2009; Hansen 2013). A growing share of credibly nonzero alignment and opposition weights involve Al-Shabaab and government forces, consistent with a shift from a fragmented system to one where a smaller number of actors occupy consistently consequential positions in the estimated propagation structure.

[13]Some norm-based summaries remain uncertain in late periods because the dominant pathways continue to evolve; the reorganization concerns which actor pairs carry the strongest weights, not a uniform tightening of uncertainty.

The co-belligerence networks present a stark contrast to these conflict dynamics. Figure 3 displays the co-belligerence influence weights for a comparable set of calendar quarters in which co-belligerence is observed.

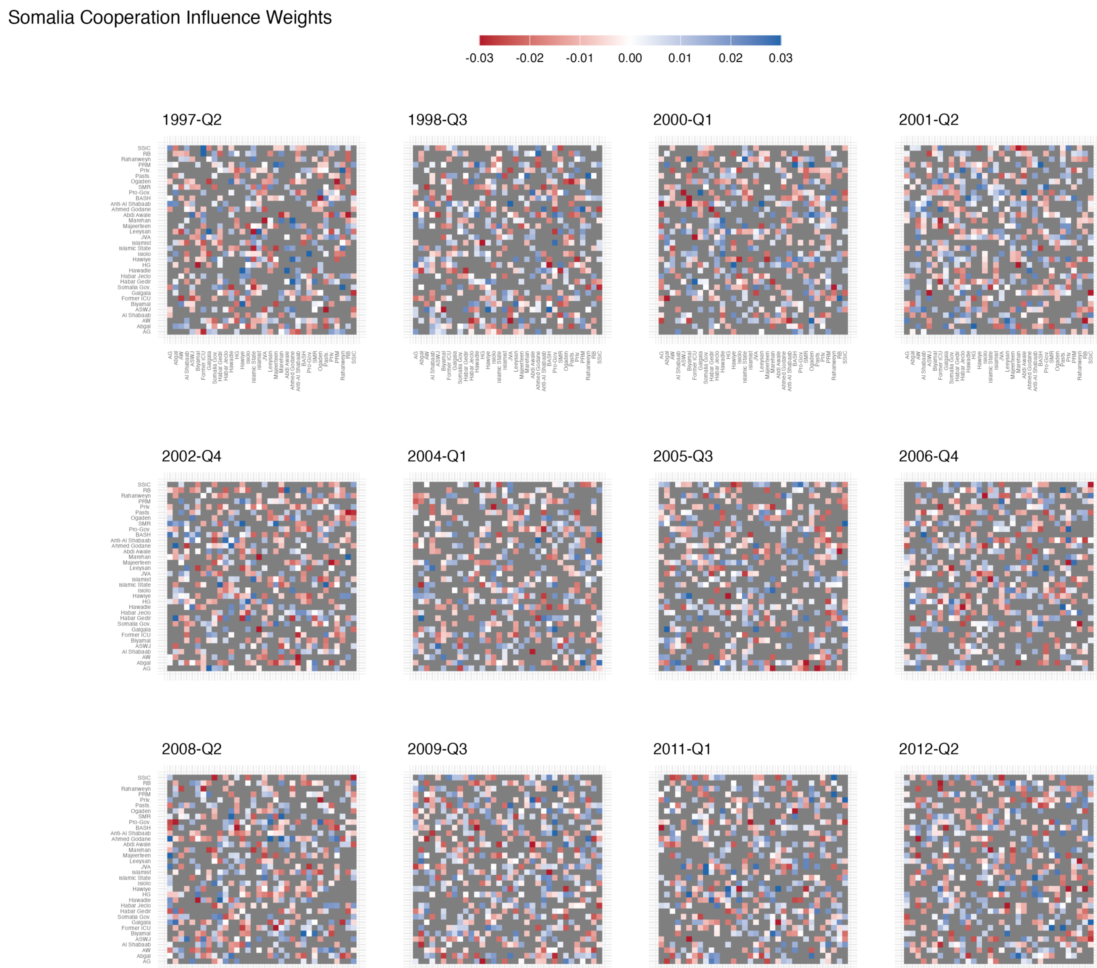

Figure 3: Cooperation influence weights $W_t = A_t B_t^\top$ at selected quarters. Color scale truncated to $[-0.03, 0.03]$; gray cells mark weights with 95% credible intervals containing zero.

The co-belligerence heat maps are dominated by gray cells, with only occasional and sporadic nonzero entries. Figure 4 reinforces this: co-belligerence weight distributions remain nearly identical across all time periods, confined to a narrow band around zero with no detectable temporal evolution. By every metric the model provides (coupling norms, dispersion of influence weights, and the random-walk innovation variance τ_A^2), conflict networks show substantially larger values than cooperation networks. The cooperation network is sparser and shorter (62 versus 88 quarters), which could reduce statistical power, but the model recovers clear time-variation in the conflict network with comparable actor counts and identical sampler settings. This asymmetry is consistent with our differential persistence expectation: co-belligerence in Somalia, as captured by ACLED-coded joint participation in battle events, appears driven by immediate tactical necessity rather than by durable strategic commitments that shape future behavior, while hostile interactions create the kind of persistent lag dependence that the model is designed to detect (Christia 2012).[14]

[14]This finding speaks to observed co-belligerence as captured in ACLED battle data, not to all forms of alliance commitment.

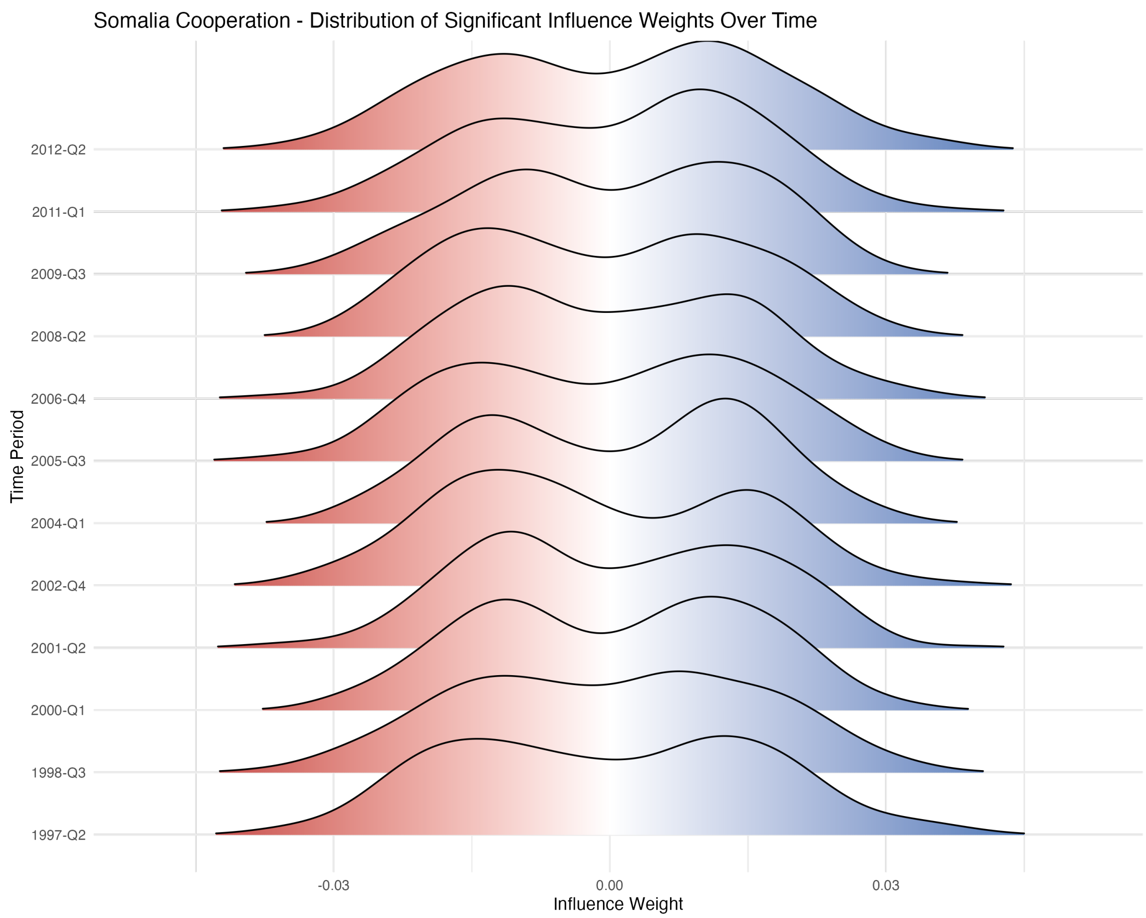

Figure 4: Cooperation: ridge plot of credibly nonzero influence weights over time.

Given these contrasting dynamics, our subsequent analyses focus on conflict networks, where we can trace how specific actors develop their structural positions and whether coupling translates into systemic leverage. We proceed in two steps: first examining actor-level coupling trajectories to test the concentration expectation, then computing propagation profiles to assess which actors' escalations are associated with the broadest and most consistent downstream effects.

4.4 Concentration of Dynamic Coupling

To examine how individual actors' structural positions evolved over time, we analyze actor-level coupling to the lag mapping. For each actor i and time period t, the dynamic bilinear model provides a posterior sample of the factor row $A_{t,i\cdot}$. This row captures actor i's responsiveness profile: the vector of weights determining how much yesterday's activity by every other actor feeds into conflict ties involving i today. Taking the Euclidean norm,

$$\text{Coupling}_i(t) = \|A_{t,i\cdot}\|_2 = \sqrt{\sum_{\ell=1}^{n} a_{i,\ell,t}^2},$$

collapses this profile into a scalar that measures how strongly actor i's next-period involvement is conditioned by the previous period's network state. We report row norms of A_t throughout; because ties are symmetric, row norms of B_t yield substantively similar patterns, and the fully scale-invariant row norms of $W_t = A_t B_t^\top$ produce the same actor ranking and temporal trajectories (see Section 3.2 for details on identification).[15]

Importantly, this measure captures latent structural coupling rather than raw event counts. Coupling need not track raw activity levels. An actor can be frequently involved

[15] In the directed case, row norms of A_t and B_t would distinguish sender responsiveness from receiver leverage. For undirected data, the two matrices play exchangeable roles and we report A_t-based summaries by convention.

in conflict yet weakly coupled if its involvement does not follow stable lag-dependent patterns in the network, for example if it enters and exits fighting in ways that are not systematically predictable from the prior quarter's configuration of interactions. Conversely, an actor can be less active in absolute terms yet strongly coupled if its participation is reliably triggered by the same lagged patterns, for example if it consistently becomes involved following escalation by a particular set of rivals or following shifts in the broader network that repeatedly precede its entry. Coupling summarizes the predictability of an actor's next-period involvement from the lag mapping, and the propagation profiles in Section 4.5 then assess whether escalations involving that actor are associated with large predicted downstream spillovers across the wider network.[16]

Figure 5 displays the group-level influence trajectories for four focal actors. The graphs include the posterior median (blue line) and 95% credible intervals (gray ribbon) across 88 quarterly time periods. Wide ribbons indicate periods where an actor's influence is uncertain from the data; narrow ribbons reveal stable roles.

[16]We use the L2-norm because it provides a rotation-invariant summary that enables comparison across time periods. Unlike raw event counts or simple centrality measures, it captures an actor's embedding in the autoregressive structure of the network.

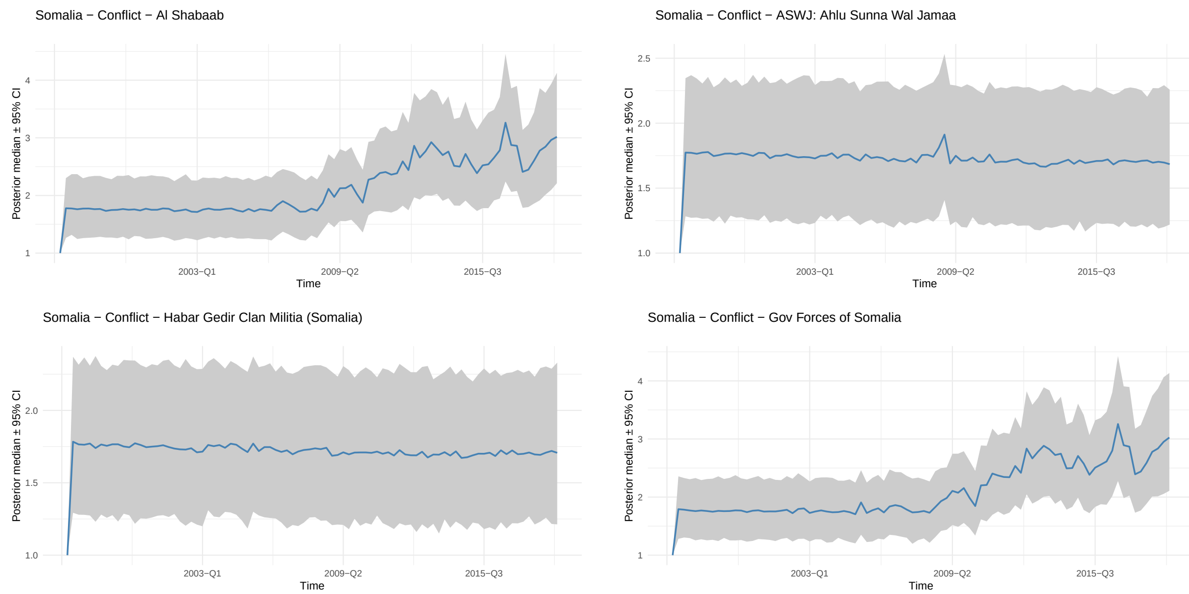

Figure 5: Somalia conflict: actor coupling trajectories ($\|A_{t,i\cdot}\|_2$).

The coupling trajectories from Figure 5 indicate a shift from a multi-actor conflict to a system increasingly organized around two principal groups, providing support for our concentration expectation, though the overall degree of concentration remains modest. The government's and Al-Shabaab's coupling norms rise steadily over the period, and their posterior median trajectories track each other closely across the full sample. This co-movement reflects a pattern in which government and insurgent forces become structurally interdependent: the government becomes more consequential in the lag structure precisely because Al-Shabaab does, and vice versa, as the conflict polarizes around a dominant axis. The pattern is consistent with the strategic interaction between insurgent and counterinsurgent forces documented in the civil war literature (Kalyvas 2006; Staniland 2021); as each side adapts to the other, their mutual engagement increasingly shapes the broader network's lagged dependence structure. This co-evolution is a signature of what Kalyvas (2006) describes as the "joint" nature of civil war violence: the government's strategic calculus is increasingly defined by Al-

Shabaab's actions, and vice versa, producing a mutual lock-in that marginalizes other actors' structural importance even when those actors remain active combatants.

ASWJ and Habar Gedir, by contrast, remain relatively stable and lower in magnitude. Across all 25 actors, coupling becomes modestly but systematically more concentrated over time: the coupling-norm Gini rises from about 0.08 early in the sample to about 0.11 late (Appendix, Figure A12). In a sparse, multi-actor conflict network where most actors' coupling norms remain near zero, even modest increases in concentration indicate that the same few actors are increasingly responsible for the portion of the lag mapping that is stably identified across periods. The same pattern appears in the raw trajectories: the top two actors' share of total coupling rises steadily while the remaining actors stay flat (Appendix, Figure A12).

Posterior uncertainty also varies over time. Credible intervals widen for Al-Shabaab and government forces in later periods, indicating that while both actors are consistently estimated to become more tightly coupled, the precise magnitude of that coupling is less precisely pinned down in some late-sample quarters. This is consistent with a setting in which influence structure is both concentrated and evolving, so that small changes in the fitted lag mapping can translate into noticeable changes in norm-based summaries. By contrast, the flat and well-identified trajectories of ASWJ and Habar Gedir reflect their consistent but circumscribed roles: the model estimates their coupling with high precision precisely because it is modest and stable. Whether this rising coupling translates into systemic leverage, with escalations involving the top-coupled actors generating the broadest downstream effects, is the question we turn to next.

4.5 Systemic Leverage and Cascade Consistency

The coupling analysis shows how structural embeddedness evolved, but high coupling does not guarantee that an actor's escalations will propagate widely. To assess systemic leverage, we compute period-specific propagation profiles that trace how a standardized bilateral perturbation is transmitted through the network under the fitted conditional mean dynamics. Unlike static network statistics, propagation profiles separate actors whose involvement routinely predicts network-wide changes from those whose involvement remains localized, and they capture how the same perturbation would propagate differently across conflict phases.

For each focal actor k, we perturb the latent intensity of the $k \leftrightarrow$ Government dyad by one unit at a chosen quarter τ. We choose the government as the perturbation partner because it is the one actor that engages with nearly all non-state groups, making it a natural common reference point for comparing propagation across different focal actors; the appendix verifies that the main findings are robust to using each actor's modal opponent instead. Because the government dyad is the same reference perturbation for every focal actor, differences in downstream propagation reflect differences in how the fitted lag operator reweights each actor's escalations, not differences in the choice of perturbation partner. We use Q3 2005 as an illustrative perturbation time because it falls before the Islamic Courts Union's military consolidation in Mogadishu and before the conflict polarizes around the Al-Shabaab-government axis that dominates later years. In this period, armed contention still reflects a fragmented configuration of clan-based factional competition alongside the expanding influence of Islamic courts (Barnes and Hassan 2007), rather than the more consolidated insurgent-counterinsurgent structure that follows. This makes Q3 2005 a substantively meaningful baseline for illustrating propagation profiles in a pre-consolidation phase, and the consistency analysis below repeats the calculation across a grid of perturbation times spanning the full sample. Results scale linearly in perturbation size.

We then compute the model-implied propagation of this standardized perturbation by iterating the estimated sequence of time-varying transition operators, applying the deterministic recursion $D_h = A_{\tau+h} D_{h-1} B_{\tau+h}^{\top}$ at each step. Conditional on a given posterior draw of $\{A_t, B_t\}$, this mapping is purely algebraic, and uncertainty in the propagation profiles comes from posterior uncertainty about the time-varying influence matrices rather than from forward simulation noise. At each future horizon h, we summarize the implied system-wide change by the difference in mean off-diagonal latent intensity:

$$\Delta_h(\tau, k) = \overline{\Theta_{\tau+h}^{\text{pert}}} - \overline{\Theta_{\tau+h}}, \quad \text{where } \overline{X} = \frac{1}{n(n-1)} \sum_{i \neq j} X_{ij},$$

and the sum includes the originally perturbed dyad.[17] Positive values indicate that the perturbation is associated with an increase in mean network-wide latent intensity under the fitted dynamics, while negative values indicate a decrease. We interpret large absolute changes as stronger propagation, regardless of sign. These are retrospective model-implied diagnostics: we propagate the perturbation using the estimated post-τ operator sequence $\{(A_{\tau+h}, B_{\tau+h})\}_{h \geq 1}$, so the profiles describe how the system's fitted dynamics transmit a standardized perturbation in the environment that actually followed τ.

Figure 6 reveals how different bilateral escalations propagate. For Al-Shabaab, the immediate effect (H1) is small, indicating the network does not respond instantly. By

[17]The perturbed tie's contribution is small relative to the $n(n-1)$ off-diagonal entries and does not qualitatively affect comparisons across actors. Multiplying $\Delta_h(\tau, k)$ by $n(n-1)$ gives the implied change summed across all directed dyads on the latent scale.

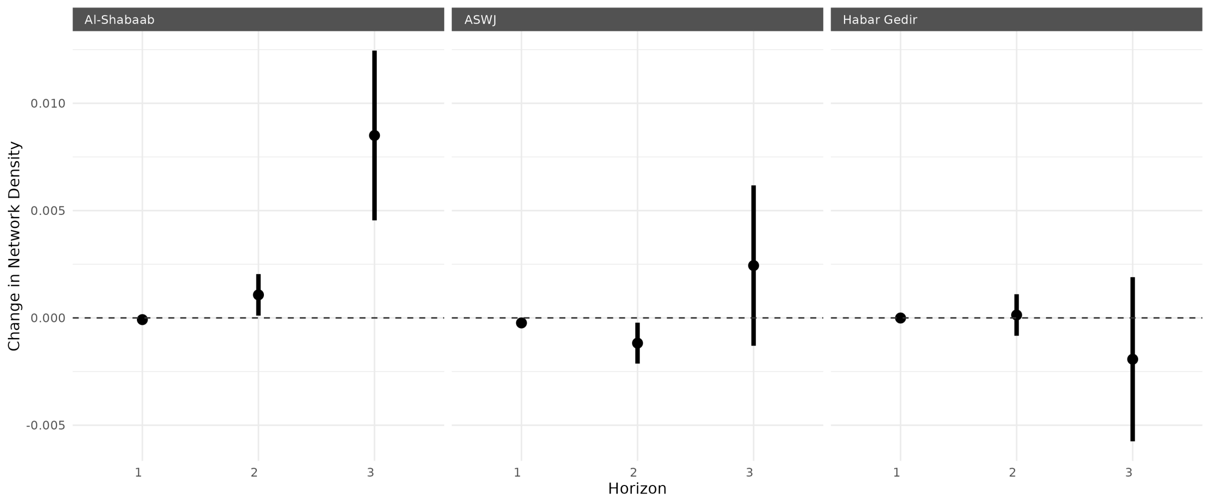

Figure 6: Propagation profiles from a single dyad perturbation at quarter 35 (Q3 2005). Each panel shows downstream changes when that actor's tie with the government is increased by one unit. The x-axis shows horizons: H1 (next quarter), H2 (two quarters ahead), H3 (three quarters ahead). The y-axis shows the change in mean off-diagonal conflict intensity. Points show posterior means; vertical bars capture posterior uncertainty across MCMC draws.

H2, we see modest positive effects emerging. By H3, the effect has grown substantially. This delayed growth in absolute magnitude indicates multi-step propagation under the fitted dynamics: the standardized increase is reweighted onto additional dyads over successive periods. Several mechanisms could generate such a pattern in the observed event record. Delayed retaliation is one possibility: an Al-Shabaab escalation triggers government response, which in turn provokes reactions from clan militias aligned with or opposed to either side. Third-party activation is another: peripheral actors who were not directly involved in the initial escalation adjust their behavior in subsequent quarters as the local balance of power shifts. The diagnostic itself is agnostic about which pathway dominates, but the multi-quarter build-up is consistent with a propagation regime in which bilateral violence feeds through intermediary actors before reaching its full system-wide extent.

ASWJ and Habar Gedir show limited and uncertain propagation. For ASWJ, the posterior mean spillover fluctuates across horizons and is estimated imprecisely, which indicates that the model does not recover a stable, system-wide propagation signature for escalations involving this actor. For Habar Gedir, estimated spillovers remain near zero across horizons, with a slight tendency toward dampening at longer horizons. These patterns are consistent with involvement that is more context-dependent and less reliably embedded in network-wide lag dynamics. ASWJ, for instance, is commonly described as a locally grounded, Sufi-aligned armed actor whose salient wartime role is tied to resisting Al-Shabaab in specific central regions (Felbab-Brown 2020); its engagements, while important in those arenas, may be less consistently connected to system-wide reweighting of conflict ties. Habar Gedir, which we treat as a clan-based militia actor in this dataset, plausibly captures a form of conflict involvement that is more localized and contingent, relative to actors embedded in the dominant insurgent-counterinsurgent axis. Both cases suggest that raw participation in violence is insufficient for systemic leverage; what matters is whether an actor's engagements are positioned to activate lag-dependent responses from multiple other actors.[18] The contrast with Al-Shabaab is that Al-Shabaab-linked escalations are associated with a clearer multi-step build-up in predicted spillovers over successive quarters, consistent with a propagation regime in which third-party activation and delayed responses are more systematically encoded in the lag mapping.

The propagation profiles above capture dynamics at one specific time point. To assess whether these patterns persist across different phases of the conflict, we repeat the same perturbation over a grid of perturbation times spanning the sample and horizons H1 through H3. For each actor, we summarize stability with a coefficient of variation computed over the posterior mean propagation effects across perturbation times and horizons, defined as the standard deviation divided by the mean absolute effect.

[18] Sparser observation of some dyads and the coarsening of events to quarters can also contribute to imprecise propagation estimates for less active actors.

Because this statistic can still be large when average effects are very small, we interpret higher values as indicating weak and/or context-dependent propagation rather than instability alone.

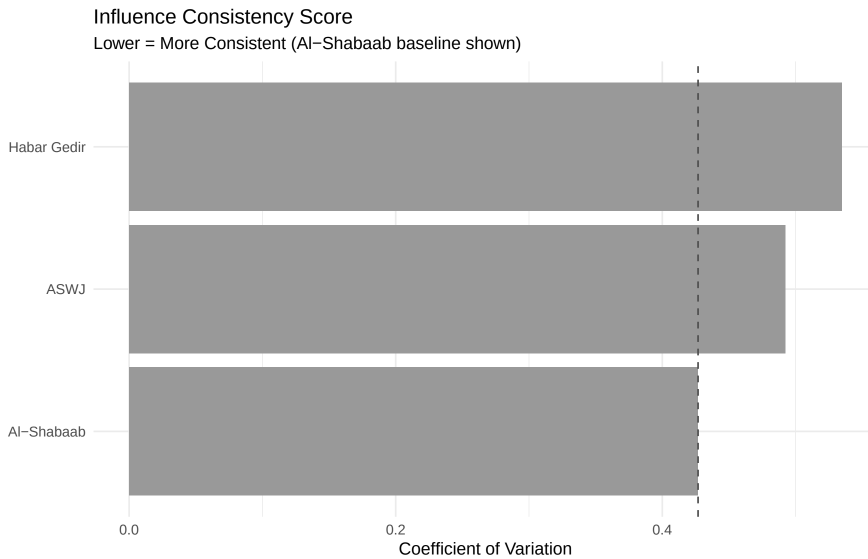

Figure 7: Influence consistency across perturbation times. Lower coefficient of variation indicates more stable, predictable propagation effects regardless of when the perturbation occurs. The dashed line shows Al-Shabaab's CV as a baseline for comparison.

Because the coefficient of variation measures variability relative to average absolute effect, it favors actors whose propagation is both large in magnitude and stable relative to that magnitude, which is precisely the combination we interpret as consistent systemic leverage. Figure 7 shows that Al-Shabaab combines the largest mean absolute downstream effects with the lowest variability relative to that magnitude, yielding the lowest CV and indicating the most consistent predicted propagation profile across perturbation times. This consistency is a distinctive finding: it means that Al-Shabaab's systemic leverage is not an artifact of one particular conflict phase but persists across the structural transitions from the warlord era through the ICU period and into the AMISOM counterinsurgency. These patterns complement our coupling findings by separating structural embeddedness from systemic leverage. Al-Shabaab does not just occupy a tightly coupled structural position; its involvement reliably predicts multi-step propagation through the network across different periods. Under the fitted dynamics, standardized increases involving Al-Shabaab imply larger network-wide downstream changes over the next few quarters than comparable increases involving other actors, and this pattern is more stable across time than for any other actor.

In contrast, ASWJ and Habar Gedir show limited and unpredictable propagation. Even when these actors escalate substantially with the government, the model-implied effects tend to remain localized or depend heavily on the current configuration of alliances. The direction of downstream changes is also moderately variable for all three non-state focal actors (Al-Shabaab, ASWJ, and Habar Gedir), with sign consistency ranging from 0.53 to 0.80 across horizons, reflecting the small absolute magnitude of intensity changes. Al-Shabaab's advantage is less about always pushing the network in one direction and more about being associated with consistently large predicted cascades regardless of sign. This distinction between structurally consequential and locally confined actors, identified through the consistency of predicted cascades rather than raw event counts, illustrates the type of dynamic network diagnostic that static models cannot provide.

4.6 Cross-Case Patterns

The appendix reports parallel analyses for the Democratic Republic of Congo and Sudan (25 actors each, up to the 1997-2020 period).[19] Three qualitative patterns hold across all three settings. First, conflict networks display substantially stronger lag dependence than co-belligerence networks by every diagnostic the model provides. Second, gov-

[19] Detailed cross-case results, including innovation variance comparisons, coupling trajectories for the top actors in each conflict, and variance decomposition tables, appear in the appendix (Table A2, Figure A14, Table A1).

ernment forces and the most tightly coupled non-state actor emerge as the two most structurally embedded actors in each conflict network, consistent with a tendency for protracted conflict systems to organize around a dominant government-adversary axis. Third, the time-varying latent process explains a large share of residual variance beyond the dyad-specific baseline in conflict networks but adds little in co-belligerence networks. The cross-case variation is itself informative: Somalia, where the conflict underwent the most dramatic structural reorganization, shows the largest role for time-varying dynamics, while Sudan, with a more stable government-rebel axis throughout, shows a smaller but still nontrivial contribution.

Taken together, each of our theoretical expectations finds support, though with varying degrees of strength: coupling concentrates modestly but consistently over time, conflict networks show stronger lag dependence than co-belligerence networks, and cascade heterogeneity separates systemically consequential actors from locally confined ones. A static network model would have identified Al-Shabaab as central but could not have shown when that centrality emerged, how it co-evolved with government forces, or whether it translated into consistent systemic leverage rather than merely high activity.

5 Discussion

The central empirical finding is that conflict systems can sustain substantial violence while shifting sharply in how that violence propagates across actors and over time. In Somalia, the estimated lagged dependence mapping evolved from a diffuse, multipolar structure to one increasingly organized around Al-Shabaab and government forces. This transition was not simply a matter of rising violence or greater activity by particular groups. It was a change in the conflict system's propagation structure: escalations involving Al-Shabaab became increasingly associated with multi-quarter, network-wide predicted spillovers, while escalations involving other highly active groups remained localized and contingent. This distinction between aggregate violence levels and propagation structure is difficult to detect with the aggregate outcomes and static relational summaries that dominate existing work on civil war duration and intensity (Fearon 2004; Cunningham 2013), which cannot distinguish tightly coupled conflict systems from diffuse ones with similar aggregate violence levels. The key implication is that how violence spreads is a distinct object from how much violence occurs, and the two can diverge sharply within the same conflict.

This finding has broader implications for the study of intrastate conflict. It suggests that conflict duration and persistence may operate through distinct structural channels. Two conflict systems with similar levels of aggregate violence can differ markedly in their propagation structure; one may be tightly coupled, so that localized escalation reliably spreads, while the other remains diffuse. The shift from fragmented to concentrated lag dependence that we document in Somalia represents a qualitative change in how the conflict system processes violence, and it coincides with a period widely characterized as difficult to resolve through negotiation and external stabilization efforts (Menkhaus 2007; Williams 2018). A concentrated propagation structure may be more operationally salient because a large share of the fitted lag dependence is associated with a small number of actors, suggesting that forecasting and monitoring systems that prioritize those actors may capture a disproportionate share of the periods when network-wide cascades are most likely under the fitted dynamics.[20] More broadly, this suggests that theories of civil war escalation and persistence should specify not only mechanisms of violence, but mechanisms governing how lagged interdependence it-

[20]Causal claims about what interventions would achieve require experimental or quasi-experimental designs beyond the scope of this paper.

self reorganizes over time.

The coupling/leverage decomposition also speaks to intervention design. Efforts that target the most active actors are not necessarily targeting the most systemically consequential ones. In our estimates, Al-Shabaab scores high on both coupling and leverage, but other highly active groups such as ASWJ and Habar Gedir do not. Forecasting and de-escalation efforts that ignore changing lag dependence may miss when cascades are most likely and which actors most consistently drive them. Similarly, peace negotiations that focus exclusively on the most visibly active parties may not address the actors whose involvement is most structurally embedded in the system's propagation dynamics; our framework provides a principled way to identify those actors and track whether their structural positions change in response to ceasefires, interventions, or other shocks.

Across the three settings, a recurring pattern emerges: protracted conflicts tend to organize around a dominant government-adversary axis in the lag mapping, and hostile networks exhibit stronger and more persistent lag dependence than co-belligerence networks.

Methodologically, the central contribution is to make the time-varying lag mapping an estimand that can be recovered from political network data at meaningful temporal resolution. Existing dynamic network approaches either impose time-invariant dependence parameters or focus on evolving latent similarity, neither of which directly targets the mapping from past interactions to future ones. By parameterizing that mapping as $A_t \Theta_{t-1} B_t^\top$, the model yields influence trajectories that support the actor-level and system-level diagnostics at the heart of the paper. The row-wise FFBS decomposition keeps inference tractable by exploiting conditional structure in the state-space representation, making it feasible to estimate evolving influence in the multi-actor, multi-decade conflict settings that motivate our application. The resulting framework enables analyses that were previously infeasible: dating when specific influence relationships shifted, separating coupling from leverage at the actor level, and comparing propagation regimes across conflict phases and across cases.

Several limitations should guide interpretation. The model is observational, and all diagnostics are predictive rather than causal, identifying which actors and periods are associated with the strongest model-implied cascades rather than the consequences of a hypothetical intervention. The co-belligerence measure captures only one visible form of tactical alignment, namely joint participation in battle events, and may miss other forms of cooperation. Event data are subject to measurement variation over time, so the results are best read as estimates of propagation structure in the observed event record. Our quarterly aggregation is designed to match the timescale of retaliation and coalition adjustment, but different temporal resolutions could highlight different propagation mechanisms. Network boundary decisions also matter: our exclusion of external actors such as AMISOM, Ethiopia, and Kenya means the estimated propagation structure reflects domestic dynamics only, and the entry of these external forces may have reshaped the domestic structure in ways that appear as endogenous shifts in our estimates. Within these constraints, natural extensions include low-rank influence matrices for larger networks, regime-switching innovation variances to formalize stable versus transitional periods, and covariate-linked evolution that ties changes in the lag mapping to observable shifts in territorial control, resources, or leadership (Austin, Linkletter and Wu 2013; Olivella, Pratt and Imai 2022; Minhas and Hoff 2025).

The approach generalizes to any political network where the central question is not only who interacts with whom, but when the past network becomes most predictive of the future network, and whose involvement most consistently drives that prediction. Diplomatic exchange networks, where the question is whether last year's bilateral engagements predict this year's multilateral alignments, and interstate militarized disputes, where deterrence relationships and alliance commitments produce time-varying propagation across dyads, both exhibit the kind of evolving lag dependence that this framework is designed to recover. By tracking the evolution of predictive interdependence rather than imposing a fixed dependence structure, the framework provides a way to date structural transitions, decompose actor importance, and identify the periods and actors around which political systems tighten. In doing so, it shifts attention from how much interaction occurs to how interaction spreads, a distinction that is central not only in civil war but in many domains of political life.

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Appendix

A.1 Descriptive Analysis

A.1.1 Graph Level

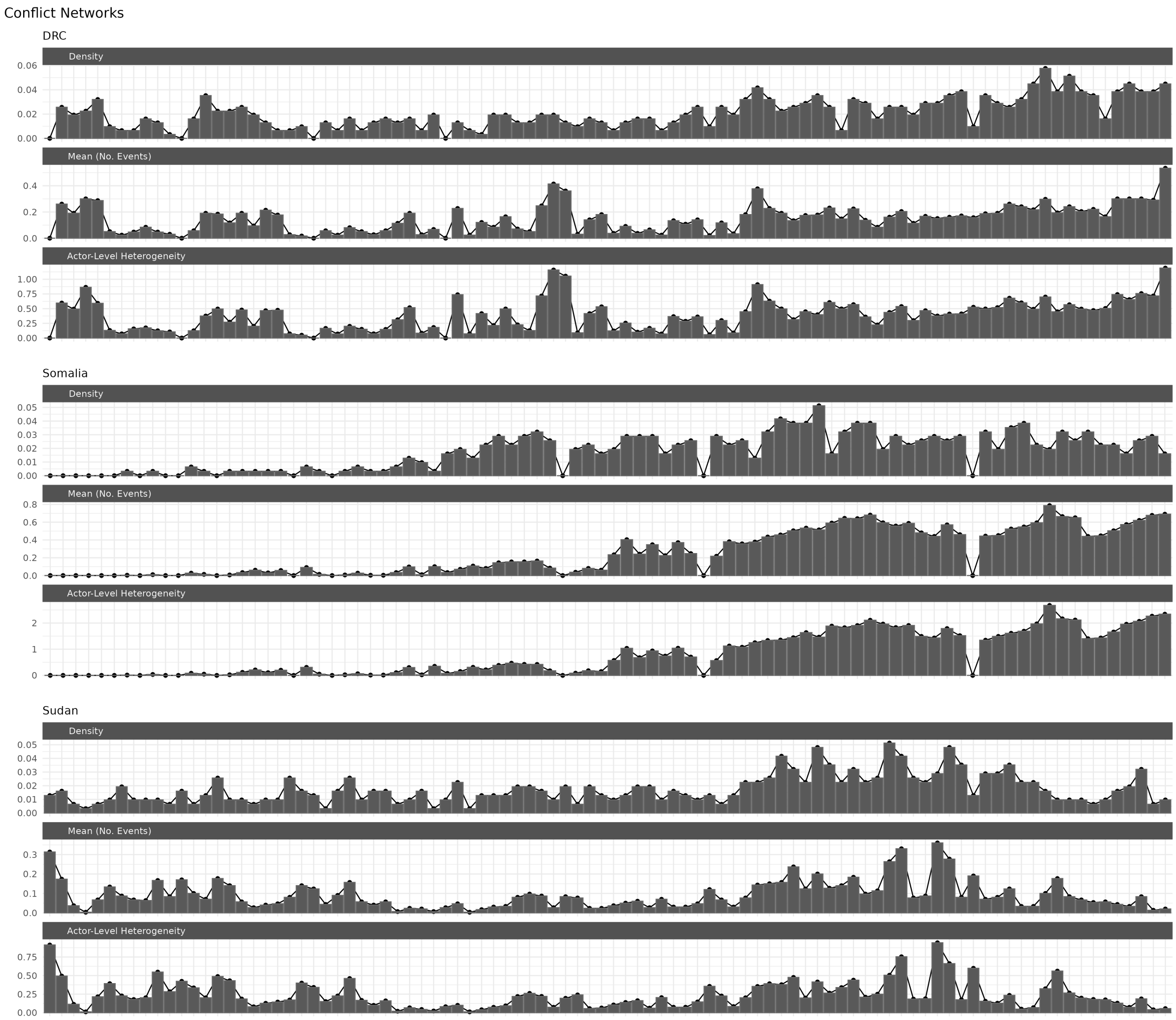

Figure A1: Conflict Network Descriptive Statistics

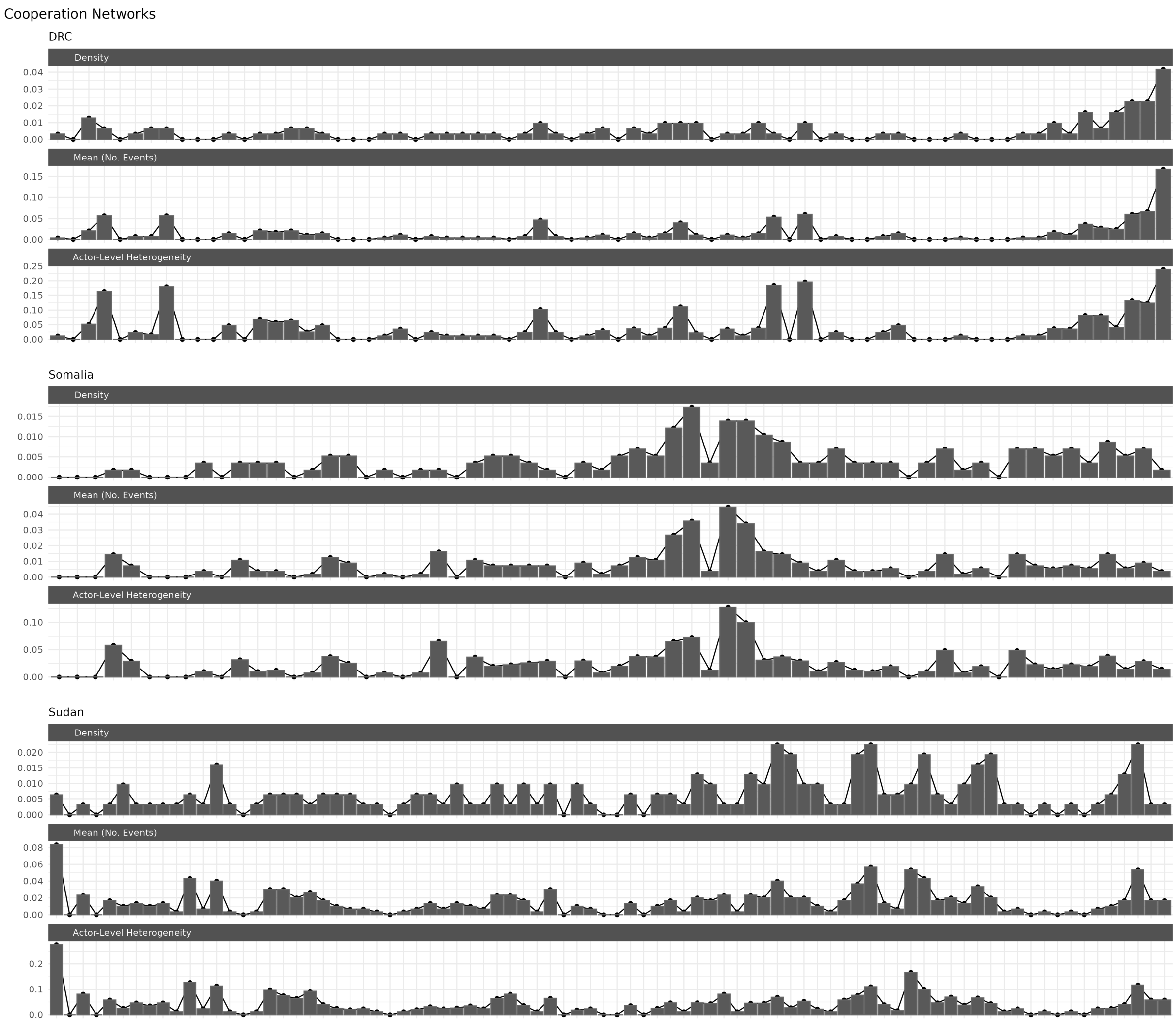

Figure A2: Cooperation Network Descriptive Statistics

A.1.2 Actor Level

Figure A3: Conflict Actor Level Network Descriptive Statistics

Figure A4: Cooperation Actor Level Network Descriptive Statistics

A.2 Trace Plots

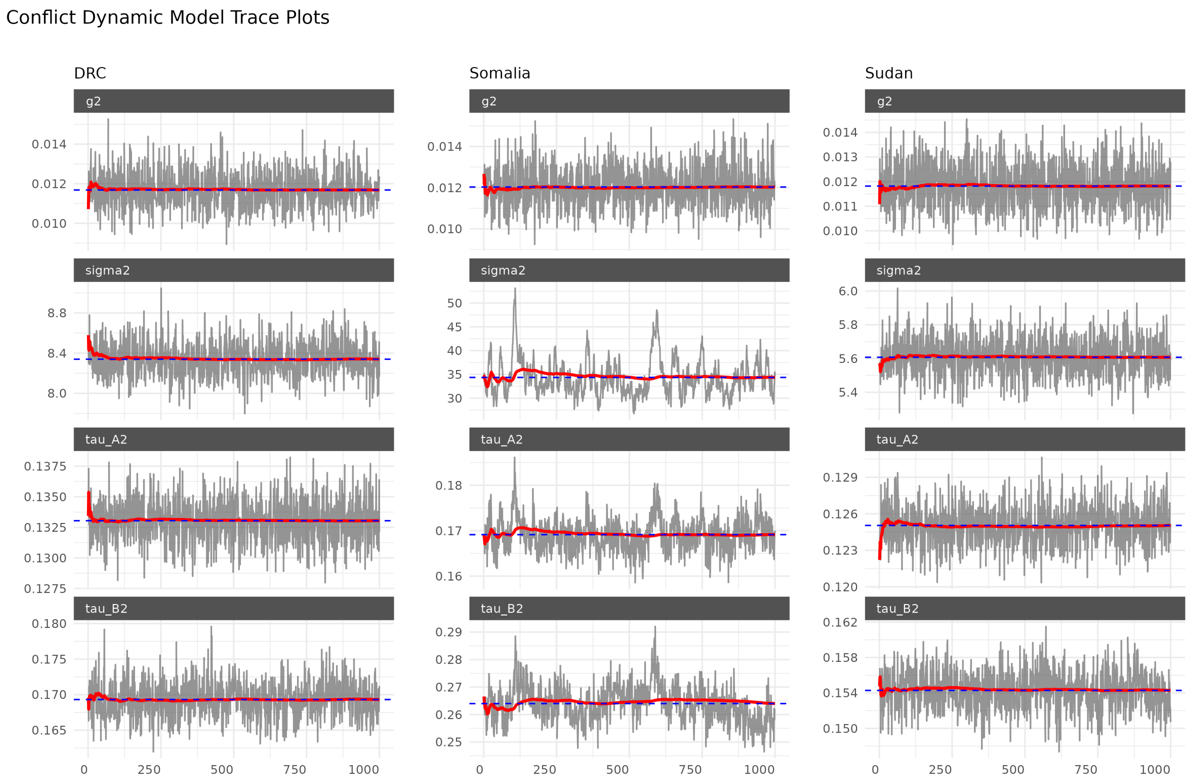

Figure A5: Conflict Dynamic Trace Plots

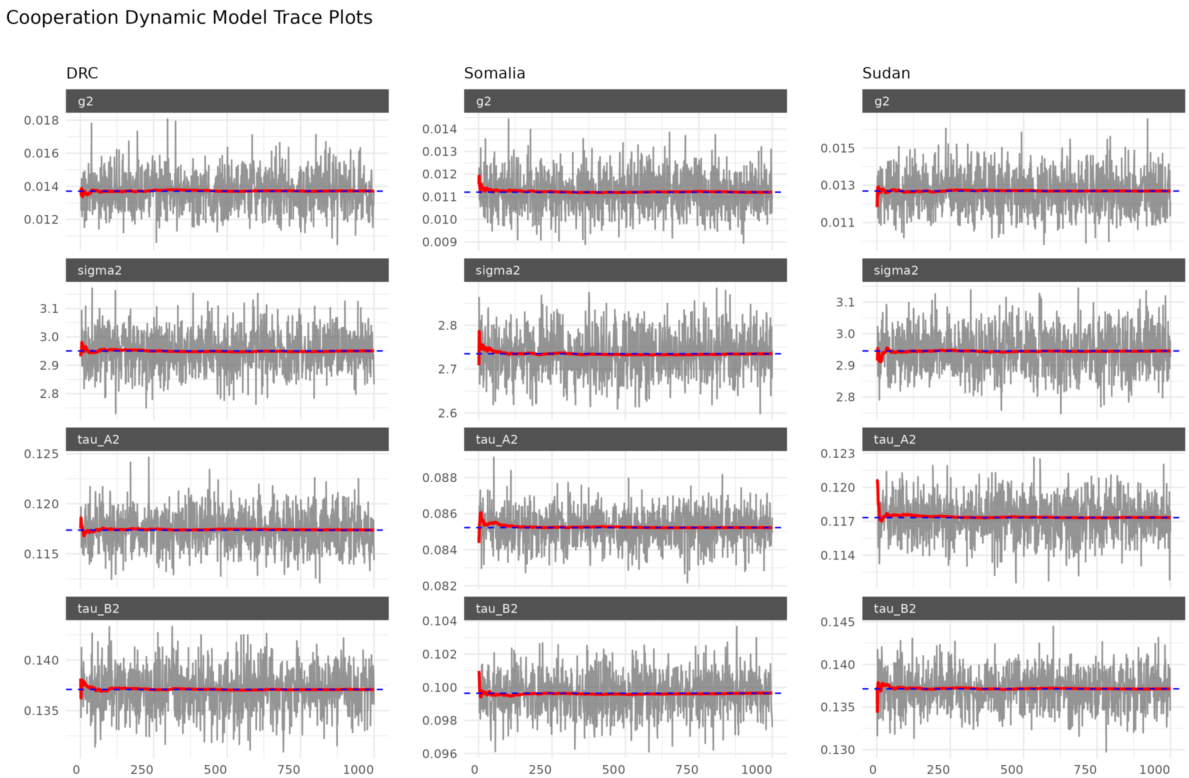

Figure A6: Cooperation Dynamic Trace Plots

A.3 Network Influence Snapshots

A.3.1 Democratic Republic of Congo

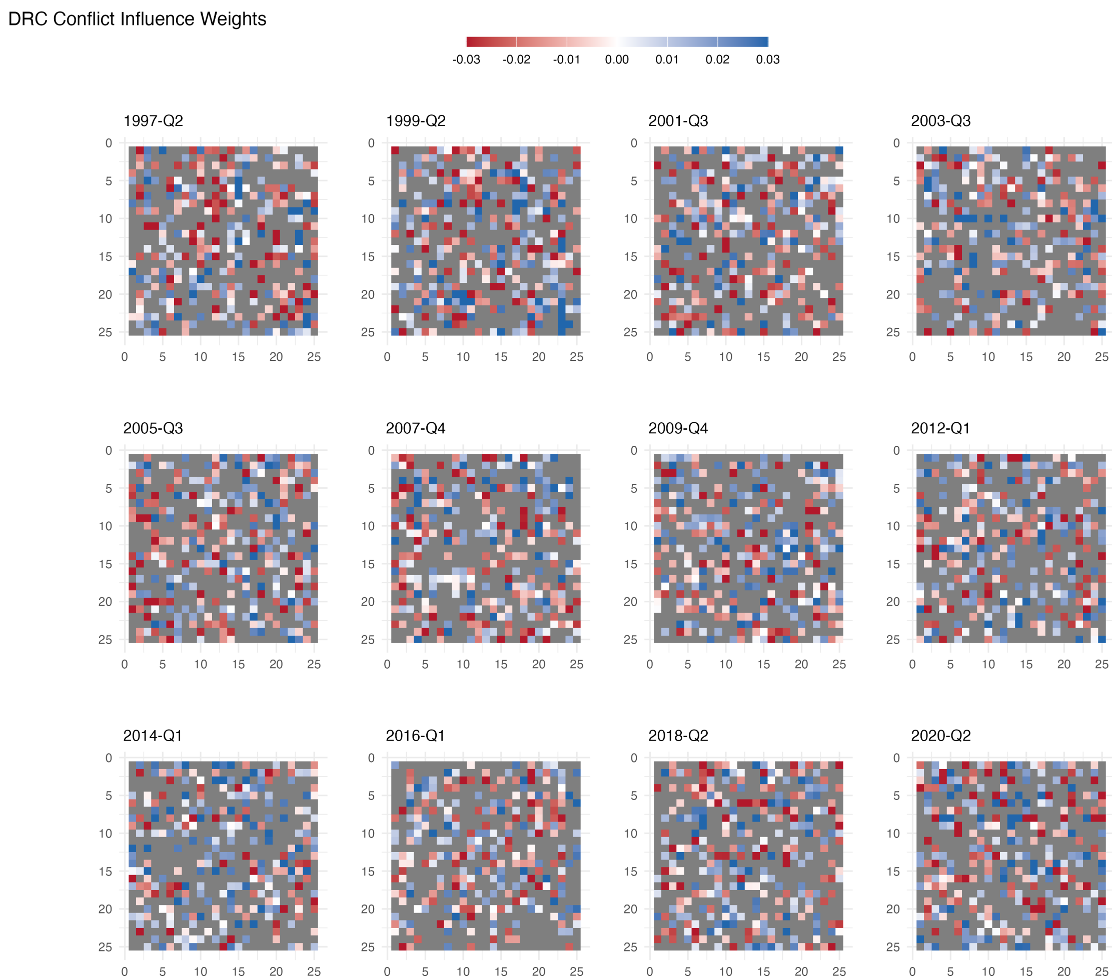

Figure A7: DRC conflict influence snapshots, gray cells indicate influence weights whose 95% credible intervals contain zero.

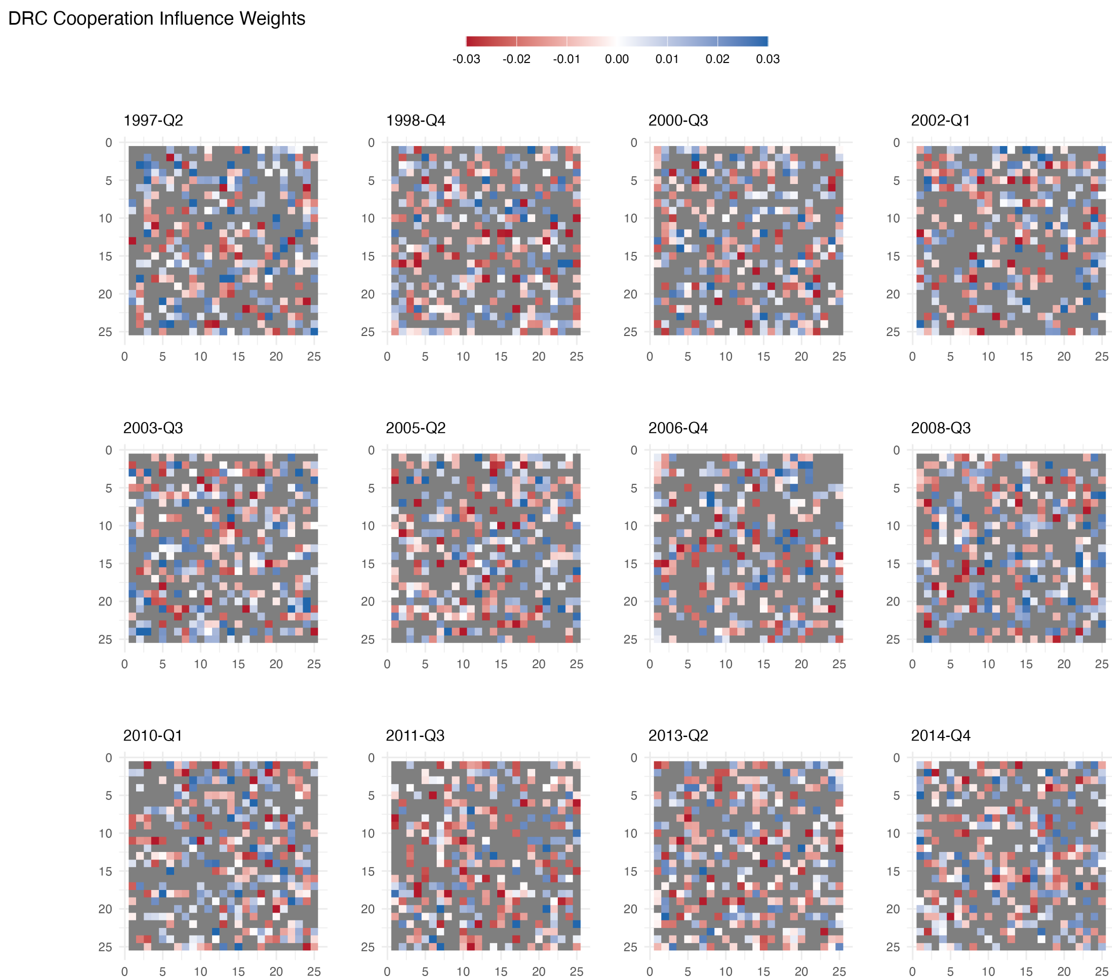

Figure A8: DRC cooperation influence snapshots, gray cells indicate influence weights whose 95% credible intervals contain zero.

A.3.2 Sudan

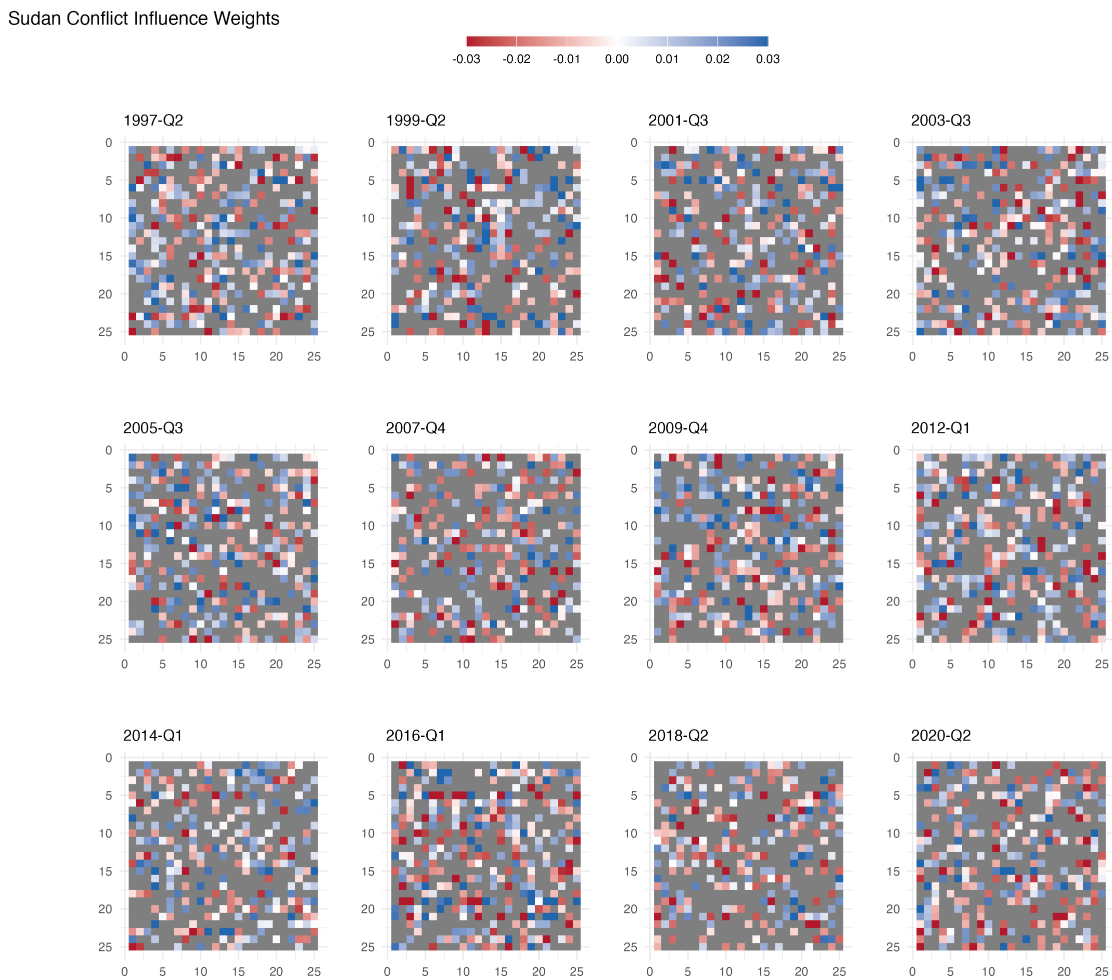

Figure A9: Sudan conflict influence snapshots, gray cells indicate influence weights whose 95% credible intervals contain zero.

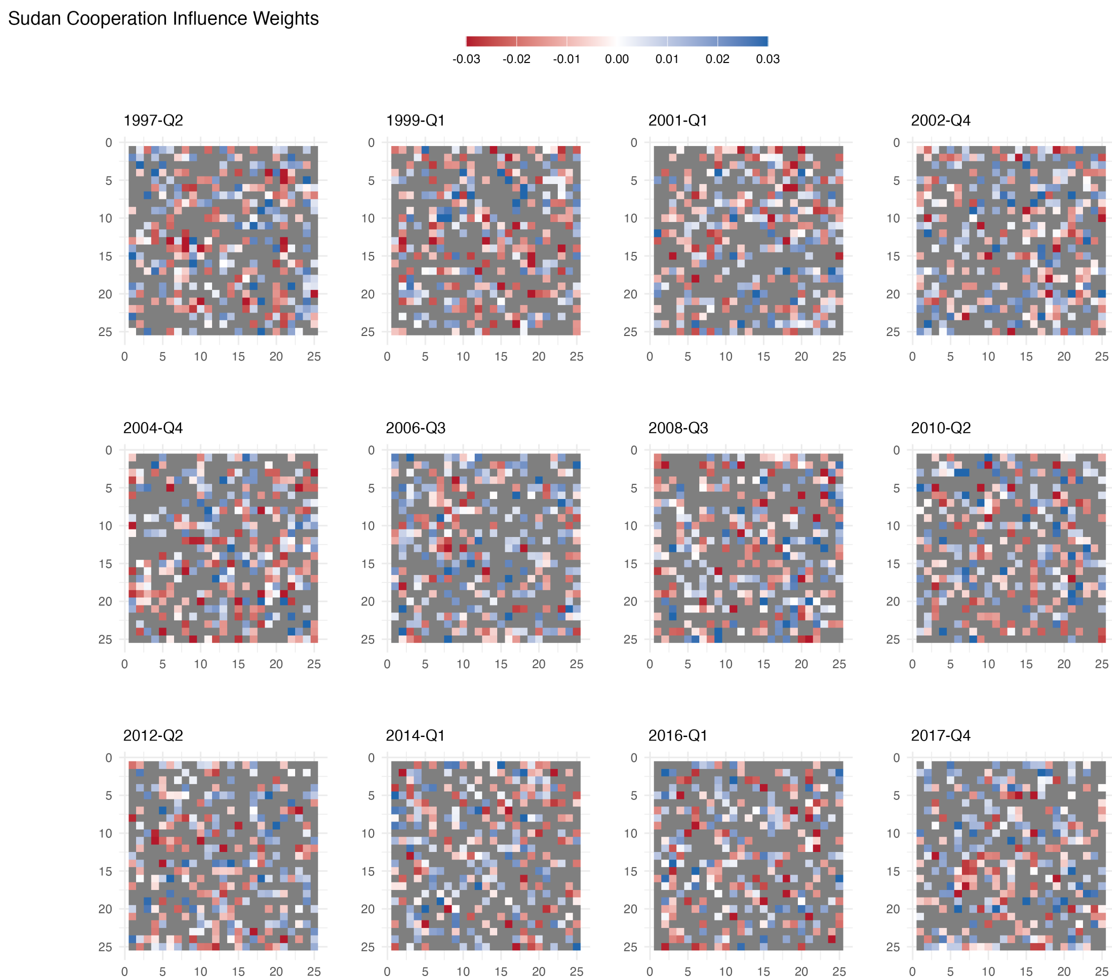

Figure A10: Sudan cooperation influence snapshots, gray cells indicate influence weights whose 95% credible intervals contain zero.

A.4 Full Conditional Distributions for the Static Model

The conjugate priors yield closed-form full conditional distributions for all parameters of the static bilinear model. The process variance σ^2 updates via an inverse-gamma distribution. Defining residuals $R_t = \Theta_t - A\Theta_{t-1}B^\top$ for $t = 2, \ldots, T$:

$$\sigma^2|\text{rest} \sim \text{IG}\left(\frac{\nu_0 + n^2(T-1)}{2}, \frac{\nu_0\sigma_0^2 + \sum_{t=2}^{T} \|R_t\|_F^2}{2}\right). \tag{A1}$$

The observation variance σ_Y^2 updates similarly using observation residuals $S_t = Y_t - M - \Theta_t$.

The baseline matrix M admits element-wise normal updates. For each entry m_{ij}, letting $\bar{y}_{ij} = T^{-1}\sum_t (y_{ij,t} - \Theta_{ij,t})$:

$$m_{ij}|\text{rest} \sim \mathcal{N}\left(\frac{T\bar{y}_{ij}}{T + \sigma_Y^2\kappa_M^{-2}}, \frac{\sigma_Y^2}{T + \sigma_Y^2\kappa_M^{-2}}\right). \tag{A2}$$

The influence matrices A and B update row-wise through Bayesian linear regression. For each row $\mathbf{a}_i$ of matrix A, we construct the regression system:

$$\mathbf{y}_i = [\Theta_{i,\cdot,2}, \ldots, \Theta_{i,\cdot,T}] \in \mathbb{R}^{1\times n(T-1)}, \tag{A3}$$

$$\mathcal{X} = [\Theta_{t-1}B^\top]_{t=2}^{T} \in \mathbb{R}^{n\times n(T-1)}, \tag{A4}$$

so that $\mathbf{y}_i = \mathbf{a}_i\mathcal{X} + \sigma\mathbf{e}_i$. The conjugate normal prior yields:

$$\mathbf{a}_i|\text{rest} \sim \mathcal{N}(\mu_A, V_A), \tag{A5}$$

where $V_A = (\sigma^{-2}\mathcal{X}\mathcal{X}^\top + \kappa_A^{-2}I_n)^{-1}$ and $\mu_A = V_A(\sigma^{-2}\mathcal{X}\mathbf{y}_i^\top)$. Matrix B updates analogously by switching sender and receiver roles. For computational efficiency, $\mathcal{X}\mathcal{X}^\top$ is pre-computed once per iteration and reused across all row updates.

A.5 Full Conditional Distributions for the Dynamic Model

The dynamic model's posterior inference follows the same conjugate structure as the static case, with the key extension that A_t and B_t are now sampled as entire trajectories via row-wise FFBS rather than as single matrices.

A.5.1 Sampling $B_{1:T}$ via Row-wise FFBS

Matrix $B_{1:T}$ updates analogously to $A_{1:T}$. Starting from $\Theta_t = A_t \Theta_{t-1} B_t^\top + \sigma E_t$, the jth column of Θ_t satisfies

$$\Theta_{\cdot j,t} = (A_t \Theta_{t-1})\, b_t^{(j)} + \sigma e_{\cdot j,t},$$

where $b_t^{(j)} = B_t[j,\cdot]^\top \in \mathbb{R}^n$.

For row j, define

$$y_t^{(j)} = \Theta_{\cdot j,t} \in \mathbb{R}^n, \tag{A6}$$

$$X_t^{(j)} = A_t \Theta_{t-1} \in \mathbb{R}^{n \times n}, \tag{A7}$$

$$b_t^{(j)} = B_t[j,\cdot]^\top \in \mathbb{R}^n. \tag{A8}$$

This yields the observation equation

$$y_t^{(j)} = X_t^{(j)} b_t^{(j)} + \varepsilon_t^{(j)}, \quad \varepsilon_t^{(j)} \sim \mathcal{N}(0, \sigma^2 I_n),$$

with the random-walk evolution

$$b_t^{(j)} = b_{t-1}^{(j)} + \omega_t^{(j)}, \quad \omega_t^{(j)} \sim \mathcal{N}(0, \tau_B^2 I_n).$$

We apply FFBS to this n-dimensional DLM for each row $j = 1, \ldots, n$.

A.5.2 Remaining Parameter Updates

The baseline matrix M retains element-wise conjugacy:

$$m_{ij}|\text{rest} \sim \mathcal{N}\left(\frac{\sum_t (Y_{ij,t} - \Theta_{ij,t})}{T + \sigma_Y^2 \kappa_M^{-2}}, \frac{\sigma_Y^2}{T + \sigma_Y^2 \kappa_M^{-2}}\right). \tag{A9}$$

The variance parameters follow inverse-gamma updates:

$$\sigma^2|\text{rest} \sim \text{IG}\left(a_\sigma + \frac{n^2(T-1)}{2}, b_\sigma + \frac{1}{2}\sum_{t=2}^{T} \|\Theta_t - A_t \Theta_{t-1} B_t^\top\|_F^2\right), \tag{A10}$$

$$\tau_A^2|\text{rest} \sim \text{IG}\left(a_A + \frac{n^2(T-1)}{2}, b_A + \frac{1}{2}\sum_{t=2}^{T} \|A_t - A_{t-1}\|_F^2\right). \tag{A11}$$

The update for τ_B^2 follows the same pattern using innovations in B_t. The observation variance σ_Y^2 updates analogously to σ^2, using observation residuals $Y_t - M - \Theta_t$.

A.6 Ordinal Extension of Bilinear Model

In the study of international relations and political networks, researchers frequently encounter relational data that is both dynamic and ordinal in nature. Examples include diplomatic exchanges ranging from hostile to cooperative, conflict events of varying intensity, and trade relationships at different levels of engagement. Traditional approaches to modeling such data often make limiting assumptions: they either collapse ordinal categories into binary indicators (losing important gradations), treat ordinal levels as continuous values (imposing unrealistic metric assumptions), or fail to capture the complex temporal dependencies that characterize international interactions.

We propose a bilinear ordinal autoregressive model that addresses these limitations through a state-space framework where the hidden process follows a bilinear autoregressive structure. This approach preserves the ordinal structure of the data by maintaining the ordered categorical nature of observations through a latent Gaussian mechanism, rather than forcing ordinal data into binary or continuous frameworks. The model decomposes temporal dependencies into interpretable sender and receiver effects through its bilinear structure, allowing researchers to distinguish between how past sending behavior influences future sending versus how being targeted affects future targeting. Through the use of rank likelihood methods, our approach scales to ordinal data with many categories, including count data, without requiring explicit specification of numerous threshold parameters. Despite the model's complexity, we develop an efficient Gibbs sampling algorithm that exploits the problem's structure to remain tractable for moderate-sized networks.

We first develop this model for directed binary actions $\{y_{i,j,t} : 1 \leq i,j \leq n, 1 \leq t \leq T\}$, where $y_{i,j,t}$ is the indicator of an action initiated by actor i with target j at time t. We then extend it to handle ordinal outcomes, which complements the dynamic specification developed in the main text.

A.6.1 Binary model foundation

For binary actions, our state-space formulation describes the observed actions $\{y_{i,j,t}\}$ in terms of pair-specific time-invariant baseline rates $\mu_{i,j}$ and a time-dependent real-valued hidden process $\{\Theta_{i,j,t}\}$. This is done via a probit link, so that the probability of the observed data given the process can be written as:

$$p(\{y_{i,j,t}\}|\{\mu_{i,j}\},\{\Theta_{i,j,t}\}) = \prod_{t=1}^{T}\prod_{i\neq j}\Phi(\mu_{i,j}+\Theta_{i,j,t})^{y_{i,j,t}}(1-\Phi(\mu_{i,j}+\Theta_{i,j,t}))^{1-y_{i,j,t}}, \tag{A12}$$

where Φ is the standard normal CDF.

The hidden process can be expressed as $\{\Theta_1,\ldots,\Theta_T\}$, where each Θ_t is a real valued $n\times n$ sociomatrix. This sequence can be viewed as a time-series of hidden real-valued relations between actors, which in turn gives rise to the observed actions $\{y_1,\ldots,y_T\}$.

A.6.2 Bilinear autoregressive structure

We model the temporal evolution of the hidden sequence using a bilinear autoregression. Following Hoff (2015) and Minhas, Hoff and Ward (2016), we model:

$$\Theta_t = A\Theta_{t-1}B^{\top} + \sigma E_t, \tag{A13}$$

where $\sigma > 0$, and $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{n\times n}$ are parameters to be estimated. To facilitate interpretation of these parameters, this model can alternatively be written in terms of the individual elements of Θ_t as follows:

$$\Theta_{i,j,t} = \sum_{i'} \sum_{j'} a_{i,i'} b_{j,j'} \Theta_{i',j',t-1} + \sigma \epsilon_{i,j,t}.$$

This decomposition reveals the distinct roles of the two parameter matrices. The parameter $a_{i,i'}$ captures the influence of actor i''s past sending behavior (across all targets) on actor i's current sending behavior. High values indicate that i tends to follow i''s sending patterns. The parameter $b_{j,j'}$ captures how being targeted affects future targeting: it measures whether actors who were targets of actions directed at j' in the past are more or less likely to target j in the present. This bilinear structure is particularly well-suited to international relations data, where the dynamics of conflict and cooperation often depend on both who is acting and who is being targeted.

A.6.3 Extension to ordinal-valued actions

While the binary model provides a foundation, many applications involve ordinal outcomes with multiple ordered categories. To extend our framework, we leverage the latent variable representation of the probit model. Recall that a binary random variable y for which $\Pr(y = 1) = \Phi(\mu + \Theta)$ can be represented as the indicator that a normal random variable z with mean $\mu + \Theta$ and unit variance exceeds zero. This allows us to express the model described by Equations A12 and A13 as:

$$\Theta_t = A\Theta_{t-1}B^\top + \sigma E_t,$$

$$Z_t = M + \Theta_t + V_t, \tag{A14}$$

$$y_{i,j,t} = 1(z_{i,j,t} > 0), \tag{A15}$$

where $\{V_1, \ldots, V_T\}$ is a sequence of matrices with independent standard normal entries.

For ordinal outcomes where the $y_{i,j,t}$'s take values in the set $\{1, \ldots, K\}$, we obtain a bilinear ordinal autoregressive model by replacing Equation A15 with:

$$y_{i,j,t} = \max\{k : z_{i,j,t} > c_k\}, \tag{A16}$$

where $c_0 = -\infty$, $c_K = \infty$, and $c_1, \ldots, c_{K-1}$ are threshold parameters. The equations A13, A14, and A16 define our proposed bilinear autoregressive model for ordinal relational data.

A.6.4 Handling high-cardinality ordinal data

In many applications, particularly those involving event counts, the number of ordinal categories K can be very large or even unknown before observing the data. For instance, conflict event datasets may record counts ranging from 0 to several hundred events between actor pairs. Explicitly estimating threshold parameters $c_1, \ldots, c_{K-1}$ becomes impractical and may introduce unnecessary complexity.

To address this challenge, we adopt the extended rank likelihood approach (Hoff 2007). Rather than modeling the thresholds explicitly, we treat them as nuisance parameters and base inference solely on the rank ordering of the observations. Under this approach, the constraint on $z_{i,j,t}$ when $y_{i,j,t} = k$ becomes:

$$\max\{z_{i',j',t'} : y_{i',j',t'} < k\} < z_{i,j,t} < \min\{z_{i',j',t'} : y_{i',j',t'} > k\}.$$

This rank-based approach naturally handles ordinal data with many categories without parameter proliferation and can accommodate count data by treating counts as ordered categories. The method remains valid even when the number of categories varies across time or dyads, and focuses inference on the ordering of interactions rather than arbitrary threshold values. This is particularly valuable in international relations applications where the precise numeric distances between ordinal levels may be less meaningful than their relative ordering, and where event counts can vary notably across different types of interactions and time periods.

A.6.5 Estimation

Bayesian estimation and inference for the model outlined above can be performed using a Gibbs sampling algorithm. Given prior distributions over model parameters, the algorithm approximates the resulting posterior distribution through iterative sampling from full conditional distributions. The Gibbs sampler exploits the conjugacy structure of the model where possible, leading to efficient updates.

A.6.5.1 Full conditional distribution of σ^2

The parameter σ^2 represents the residual variance of the bilinear autoregressive process $\Theta_1, \ldots, \Theta_T$. Writing $R_t^{(\Theta)} = \Theta_t - A\Theta_{t-1}B^\top$, the elements of $R_2^{(\Theta)}, \ldots, R_T^{(\Theta)}$ are all i.i.d. mean-zero normal random variables with variance σ^2. Given an inverse-gamma$(\frac{\nu_0}{2}, \frac{\nu_0\sigma_0^2}{2})$ prior distribution for σ^2, the full conditional distribution is inverse-gamma($\frac{\nu_1}{2}, \frac{\nu_1\sigma_1^2}{2}$), where $\nu_1 = \nu_0 + n^2 \times (T-1)$ and $\nu_1\sigma_1^2 = \nu_0\sigma_0^2 + \sum\limits_{t=2}^{T} ||R_t^{(\Theta)}||^2$.

A.6.5.2 Full conditional distribution of M

The matrix M represents the pair-specific baseline rates of action. Letting $S_t = Z_t - \Theta_t - M$, we have that $s_{i,j,1}, \ldots, s_{i,j,T}$ are i.i.d. normal with mean zero and unit variance. A normal $(0, \tau_M^2)$ prior on $m_{i,j}$ results in a normal posterior distribution with mean $\frac{1}{T+\tau_M^{-2}} \sum_{t=1}^{T} (z_{i,j,t} - \Theta_{i,j,t})$ and variance $(T + \tau_M^{-2})^{-1}$.

A.6.5.3 Full conditional distributions of A and B

Conditional on $\Theta_1, \ldots, \Theta_T$ and B, the bilinear regression model in Equation A13 becomes a (multivariate) linear regression model, where for each $t = 2, \ldots, T$, Θ_t takes the role of the outcome and $\Theta_{t-1} B^\top$ takes the role of the predictor. Assuming the elements of A are a priori i.i.d. normal with mean zero and variance τ_A^2, the full conditional distribution of A is normal with independent rows and covariance within each row given by

$$V_A = \left(\frac{1}{\sigma^2} \sum_{t=2}^{T} \Theta_{t-1} B^\top B \Theta_{t-1}^\top + \frac{I_n}{\tau_A^2} \right)^{-1}.$$

The expectation of the matrix is given by

$$E_A = \frac{1}{\sigma^2} \left(\sum_{t=2}^{T} \Theta_t B \Theta_{t-1}^\top \right) V_A.$$

The full conditional distribution of B is similar, with mean and within-row covariance given by

$$E_B = \frac{1}{\sigma^2} \left(\sum_{t=2}^{T} \Theta_t^\top A \Theta_{t-1} \right) V_B$$
$$V_B = \left(\frac{1}{\sigma^2} \sum_{t=2}^{T} \Theta_{t-1}^\top A^\top A \Theta_{t-1} + \frac{I_n}{\tau_B^2} \right)^{-1}$$

A.6.5.4 Full conditional distribution of Θ_t

For each $t = 1, \ldots, T-1$, information about Θ_t from the remaining parameters comes from $Z_t, \Theta_{t-1}, \Theta_{t+1}, A, B,$ and σ^2. Specifically, -2 times the log conditional probability density of Θ_t has the form

$$\frac{||\Theta_t - A\Theta_{t-1}B^\top||^2}{\sigma^2} + \frac{||\Theta_{t+1} - A\Theta_t B^\top||^2}{\sigma^2} + ||Z_t - M - \Theta_t||^2,$$

where we define $s_t = \text{vec}(Z_t - M)$ as the vectorization of $Z_t - M$. The precision matrix is $I_{n^2} + \sigma^{-2}[I_{n^2} + B^\top B \otimes A^\top A]$, which yields the conditional variance of the vectorization of Θ_t:

$$V_{\Theta_t} = \sigma^2 \times (I_{n^2}(\sigma^2 + 1) + (B^\top B \otimes A^\top A))^{-1},$$

where "$\otimes$" denotes the Kronecker product. The conditional expectation is given by

$$E_{\Theta_t} = V_{\Theta_t}\left((B \otimes A)\frac{\Theta_{t-1}}{\sigma^2} + (B^\top \otimes A^\top)\frac{\Theta_{t+1}}{\sigma^2} + s_t\right),$$

where s_t is the vectorization of $Z_t - M$.

The conditional distributions of Θ_1 and Θ_T require special treatment due to boundary conditions. For Θ_T:

$$E_{\Theta_T} = V_{\Theta_T}\left((B \otimes A)\frac{\Theta_{T-1}}{\sigma^2} + s_T\right), \quad V_{\Theta_T} = \frac{\sigma^2}{\sigma^2 + 1} I_{n^2}.$$

For Θ_1, we specify a diffuse prior distribution with entries i.i.d. normal with mean zero and variance τ_Θ^2. The resulting posterior distribution has mean and variance:

$$E_{\Theta_1} = V_{\Theta_1}\left(s_1 + (B^\top \otimes A^\top)\frac{\Theta_2}{\sigma^2}\right), \quad V_{\Theta_1} = \left(I_{n^2}\left(1 + \frac{1}{\tau_\Theta^2}\right) + \frac{(B^\top B \otimes A^\top A)}{\sigma^2}\right)^{-1}.$$

Note that the covariance matrices given above are of dimension $n^2 \times n^2$. Unless n is small, it is not advisable to construct and invert these matrices directly. Instead, the computations required to sample from these multivariate normal distributions can be carried out in terms of operations on matrices of dimension $n \times n$.

A.6.5.5 Full conditional distribution of Z_t

Given values of all other parameters, the elements of each Z_t are normally distributed according to Equation A14, but constrained by the observed actions y according to Equation A16.

For the threshold approach, if $y_{i,j,t} = k$, then the full conditional distribution of $z_{i,j,t}$ is a truncated normal distribution with mean $m_{i,j} + \Theta_{i,j,t}$, variance 1, and support $(c_{k-1}, c_k]$.

For the rank likelihood approach, the constraint becomes:

$$\max\{z_{i',j',t'} : y_{i',j',t'} < y_{i,j,t}\} < z_{i,j,t} < \min\{z_{i',j',t'} : y_{i',j',t'} > y_{i,j,t}\}.$$

This ensures that the ordering of the latent variables matches the ordering of the observed outcomes.

A.6.6 Summary of posterior approximation algorithm

To summarize, a Gibbs sampler that approximates the posterior distribution of the model parameters proceeds by iterating the following steps:

Given current values $\{\sigma^2, A, B, M\}$ and $\{(Z_t, \Theta_t), t = 1, \ldots, T\}$:

1. Draw σ^2 from its inverse-gamma full conditional distribution
2. Draw the entries of M from their normal full conditional distributions
3. Draw A from its multivariate normal full conditional distribution
4. Draw B from its multivariate normal full conditional distribution

5. For each $t = 1, \ldots, T$:
   - Sample Θ_t from its multivariate normal full conditional distribution
6. For each observed level k of the ordinal outcomes:
   - Sample all $z_{i,j,t}$ for which $y_{i,j,t} = k$ from their constrained normal distributions

The ordering in step 6 is computationally efficient because all $z_{i,j,t}$ values corresponding to the same observed level share the same constraints.

A.6.7 Computational considerations

While the time-varying extension increases the parameter space from $O(n^2)$ to $O(n^2T)$, several features maintain computational feasibility. Given $\Theta_{1:T}$ and $B_{1:T}$, rows of A_t evolve as independent DLMs (and symmetrically for B_t), so each row can be updated separately. FFBS provides joint updates of entire parameter paths, avoiding slow element-wise Gibbs updates. The Kronecker identity $F_t = B_t \otimes A_t$ avoids forming large matrices explicitly.

Because the time-varying influence matrices are high-dimensional (n^2 parameters per time point for each of A_t and B_t), a central concern is overfitting. Identification of time variation comes from systematic changes in which dyads move together over time, net of time-invariant dyadic baselines. Temporal smoothing serves as regularization: the random-walk innovation variances τ_A^2 and τ_B^2 govern how quickly influence can change, and the posterior concentrates on small innovations unless the data support sharper reorganization. When τ_A^2 is small, the model borrows strength across adjacent time periods, effectively shrinking A_t toward A_{t-1}, and similarly for B_t. When τ_A^2 is larger, the data support more rapid structural change. In the empirical application, estimated innovations $\|A_t - A_{t-1}\|_F$ are small in most quarters, with larger shifts concentrated in historically plausible transition periods.

A.7 Propagation Profile Derivations

A.7.1 Network propagation diagnostics for the dynamic bilinear model

The procedure below is analogous to impulse response analysis in time-series econometrics, adapted to the bilinear network setting. We use “propagation profile” rather than “impulse response function” to emphasize the predictive, non-causal interpretation.

Let $\Theta_t \in \mathbb{R}^{n\times n}$ denote the latent network state in the dynamic bilinear model,

$$\Theta_t = A_t\,\Theta_{t-1}\,B_t^\top + \sigma E_t, \quad E_t \overset{\text{i.i.d.}}{\sim} \mathsf{MN}_{n,n}(0, I_n, I_n),$$

with observation equation

$$Y_t = M + \Theta_t + V_t, \quad V_t \overset{\text{i.i.d.}}{\sim} \mathsf{MN}_{n,n}(0, \sigma_Y^2 I_n, I_n).$$

In our undirected application we retain both A_t and B_t (estimated independently) to preserve the fully conjugate row-wise FFBS updates; for symmetric data the two matrices learn similar structure.

To construct the perturbation-based propagation diagnostic, fix a calendar quarter τ, a focal actor k, and a target actor j (in our application, j = Government). Let $S^{(k)}$ be an $n \times n$ matrix with $S^{(k)}_{k,\mathrm{Gov}} = S^{(k)}_{\mathrm{Gov},k} = 1$ and zeros elsewhere.[21] Set the initial perturbation $D_0 = S^{(k)}$ and propagate it forward deterministically using the estimated influence matrices:

$$D_h = A_{\tau+h}\,D_{h-1}\,B_{\tau+h}^\top, \qquad h = 1, \dots, H,$$

[21] Scaling is arbitrary; effects scale linearly with the shock size by the linearity of the state evolution.

so that the shocked latent state is $\widetilde{\Theta}_{\tau+h} = \Theta_{\tau+h} + D_h$. At each horizon h we summarize the network-wide impact by the change in mean off-diagonal latent intensity:

$$\mathsf{IRF}_k(h, \tau) = \overline{D_h} = \frac{1}{n(n-1)} \sum_{i \neq j} D_{h,ij},$$

where $\overline{X}$ denotes the mean of the off-diagonal elements of X. Because M appears only in the observation equation and is time-invariant, it cancels and does not affect the IRF. The procedure is executed at every MCMC draw; we store the posterior mean and credible band.

A.7.2 Measuring Consistency

An actor whose influence is structural (rooted in enduring organization, ideology, or geography) should be associated with similar predicted cascades irrespective of when it acts. Conversely, an opportunistic or capacity-constrained group will show context-dependent propagation.

We assess how stable each actor's system-level leverage is over time by computing a coefficient of variation (CV) over the posterior mean IRFs across shock times and horizons. For each actor k, let $\widehat{\mathsf{IRF}}_k(h, \tau)$ denote the posterior mean of $\mathsf{IRF}_k(h, \tau)$, and define

$$\mathsf{CV}_k = \frac{\mathrm{sd}_{\tau,h}\Big(\widehat{\mathsf{IRF}}_k(h, \tau)\Big)}{\mathrm{mean}_{\tau,h}\Big(\big|\widehat{\mathsf{IRF}}_k(h, \tau)\big|\Big)},$$

where the statistics run over seven evenly spaced quarters $\tau \in \{20, 30, \ldots, 80\}$ and horizons $h = 0, \ldots, 3$. Lower values indicate more temporally stable leverage.

A.7.3 Sign Stability of Impulse Responses

As a complement to the coefficient of variation, which captures overall magnitude stability, we assess whether the direction of each actor's spillover effect is consistent across shock times. For each actor k and horizon h, we compute sign stability as the maximum of the fraction of shock times producing a positive posterior mean intensity change and the fraction producing a negative one:

$$\text{SignStab}_k(h) = \max\Big(\frac{1}{|\mathcal{T}|}\sum_{\tau\in\mathcal{T}} \mathbf{1}\big[\widehat{\text{IRF}}_k(h,\tau) > 0\big],\ 1 - \frac{1}{|\mathcal{T}|}\sum_{\tau\in\mathcal{T}} \mathbf{1}\big[\widehat{\text{IRF}}_k(h,\tau) > 0\big]\Big),$$

where $\mathcal{T} = \{10, 15, \ldots, 80\}$ is a grid of 15 shock times. Values near 1 indicate that the spillover consistently takes the same sign regardless of when the shock occurs, while values near 0.5 indicate that the direction of spillover depends on the specific period.

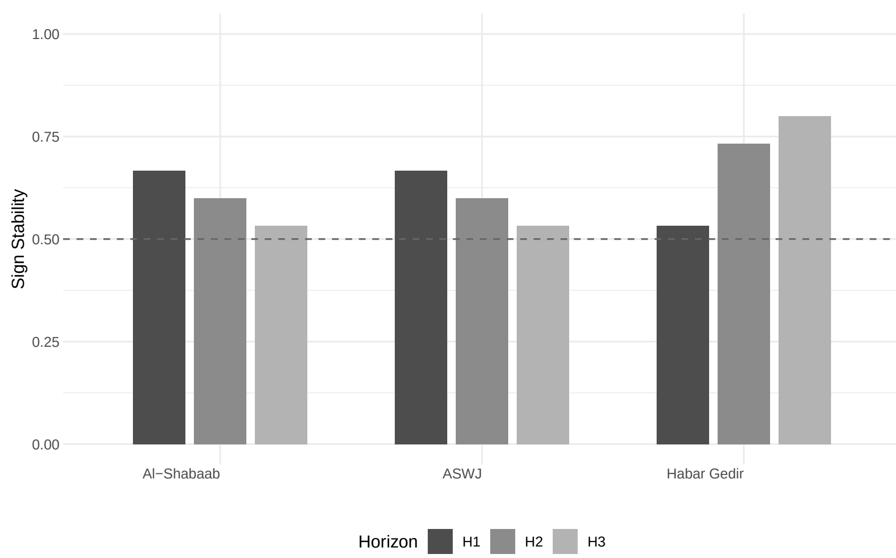

Figure A11: Sign stability of impulse responses for three focal actors, measured across 15 shock times. The dashed line at 0.5 indicates chance-level consistency (equal probability of positive and negative effects). Higher values indicate more directionally consistent spillovers.

Figure A11 shows that sign stability is moderate for all three actors (0.53-0.80), reflecting the small absolute magnitude of intensity changes at any single horizon. The most notable pattern is Habar Gedir at horizon 3, where 80% of shock times produce a negative intensity change (consistent with dampening rather than amplifying network conflict). Al-Shabaab and ASWJ both show sign stability of 0.67 at horizon 1, indicating a modest tendency toward consistent spillover direction in the immediate quarter after a shock, though this falls below the level that would indicate a robust directional signature. These results complement the CV analysis in the main text: the CV captures the ratio of variability to mean effect, while sign stability isolates directional consistency. Together, they indicate that Al-Shabaab's distinguishing feature is less about always pushing the network in one direction and more about generating consistently large (in absolute terms) and temporally stable cascades regardless of sign.

A.8 Coupling Concentration Over Time

To quantify the concentration expectation from the main text, we track the Gini coefficient of actor-level coupling norms $\|A_{t,i\cdot}\|_2$ over time. A value of 0 indicates that all actors have identical coupling norms, while values approaching 1 indicate that a single actor dominates.

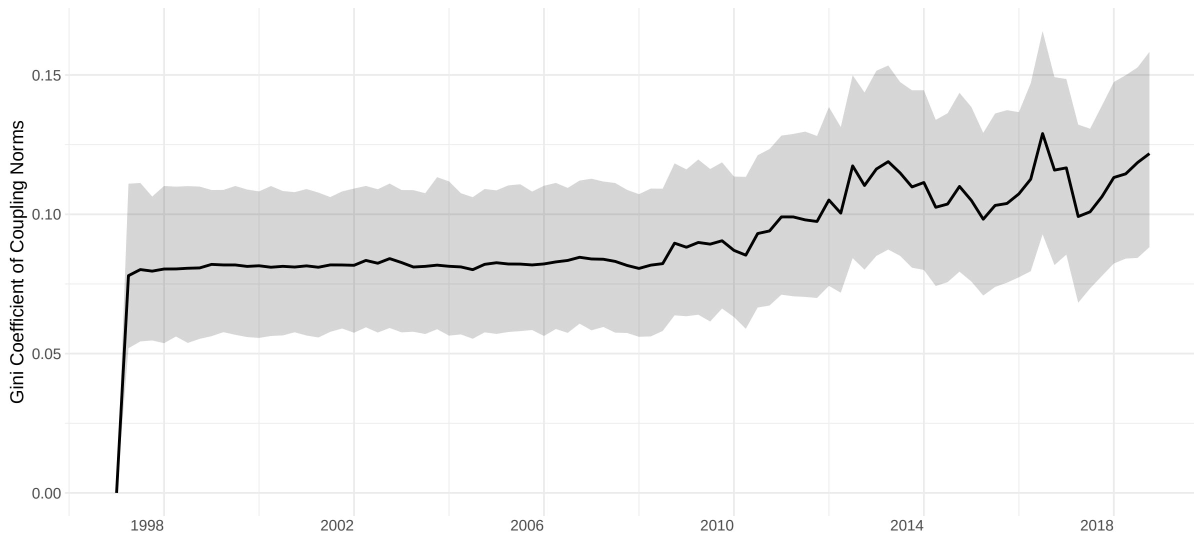

Figure A12: Gini coefficient of actor-level coupling norms over time for Somalia's conflict network. Shaded region shows pointwise 95% credible interval.

Figure A12 displays the Gini coefficient for the Somalia conflict network. The posterior mean rises from approximately 0.08 in the first third of the sample to approximately 0.11 in the final third, a roughly 40% increase, with most of the shift occurring after 2008. The absolute values remain modest (a Gini of 0.11 still indicates relatively dispersed influence across 25 actors), but the upward trend is consistent across posterior draws and aligns with the group-level trajectories in the main text.

A.9 Variance Explained by the Dynamic Latent State

To evaluate how much structure the time-varying latent process Θ_t captures, we compare the dynamic model's residual variance against an intercept-only baseline across all six country–network-type combinations.

A.9.1 Variance decomposition

Under an intercept-only baseline, the predicted network at every time point is the posterior mean intercept $\hat{M}$, so the residual variance is $\text{MSE}_{\text{baseline}} = \frac{1}{T}\sum_{t=1}^{T}\overline{(Y_t - \hat{M})^2}$, where the inner average is over all off-diagonal dyads. For the dynamic model, the observation equation $Y_t = M + \Theta_t + V_t$ implies that the only residual variance not captured by the time-varying latent state Θ_t is the observation noise σ_Y^2. Variance reduction is therefore $1 - \hat{\sigma}_Y^2/\text{MSE}_{\text{baseline}}$. This measures the share of dyadic variation that the autoregressive latent process explains beyond a time-invariant mean.

Table A1: Variance decomposition: intercept-only baseline residual variance versus dynamic observation noise $\hat{\sigma}_Y^2$. Variance reduction indicates the share of the baseline's residual that the time-varying latent state explains.

	Baseline Resid. Var.	Dynamic $\hat{\sigma}_Y^2$	Var. Reduction
Somalia Conflict	16.39	0.98	94%
DRC Conflict	2.66	0.90	66%
Sudan Conflict	1.25	0.84	33%
Somalia Cooperation	0.02	0.70	< 0[a]
DRC Cooperation	0.09	0.72	< 0[a]
Sudan Cooperation	0.08	0.72	< 0[a]

[a] $\hat{\sigma}_Y^2$ exceeds baseline residual variance; autoregressive specification adds no explanatory power.

For all three conflict networks, the dynamic latent process substantially reduces residual variance relative to the intercept-only baseline (33-94%), indicating that the autoregressive structure captures real temporal dependence. Somalia's particularly large reduction reflects the pronounced dynamics documented in the main text. For cooperation networks, the intercept alone already explains most of the signal: the baseline residual variance (0.02-0.09) is well below the dynamic model's observation noise floor ($\hat{\sigma}_Y^2 \approx 0.7$). This occurs because the dynamic model's $\hat{M}$ is estimated jointly with Θ_t, so $\hat{\sigma}_Y^2$ reflects the observation noise learned during joint estimation rather than a marginal variance decomposition. The negative variance reduction indicates that there is too little temporal variation for the autoregressive specification to improve upon the intercept-only fit. This asymmetry between conflict and cooperation is itself a key finding: it is consistent with the pattern of stronger lagged dependence in conflict than in co-belligerence reported in the main text.

A.10 Structural Dispersion Index

To complement the distributional evidence from the ridge plots in the main text, we compute a scalar structural dispersion index at each time point: the standard deviation of all off-diagonal elements of the posterior mean influence weight matrix $\hat{W}_t = \hat{A}_t \hat{B}_t^\top$. This measures heterogeneity in pairwise influence-profile alignment across the network.

Figure A13 shows the dispersion index for Somalia's conflict and cooperation networks. Conflict dispersion is consistently high (sd $\approx$ 0.8) across the entire sample period, while cooperation dispersion remains low (sd $\approx$ 0.35). The conflict index does not exhibit a strong monotonic trend; instead, it establishes a high baseline early and remains elevated. The structural evolution documented in the main text, where specific actor pairs (particularly Al-Shabaab–Government) develop increasingly strong positive or negative weights, is therefore a reorganization of the influence structure rather than a uniform increase in overall dispersion.

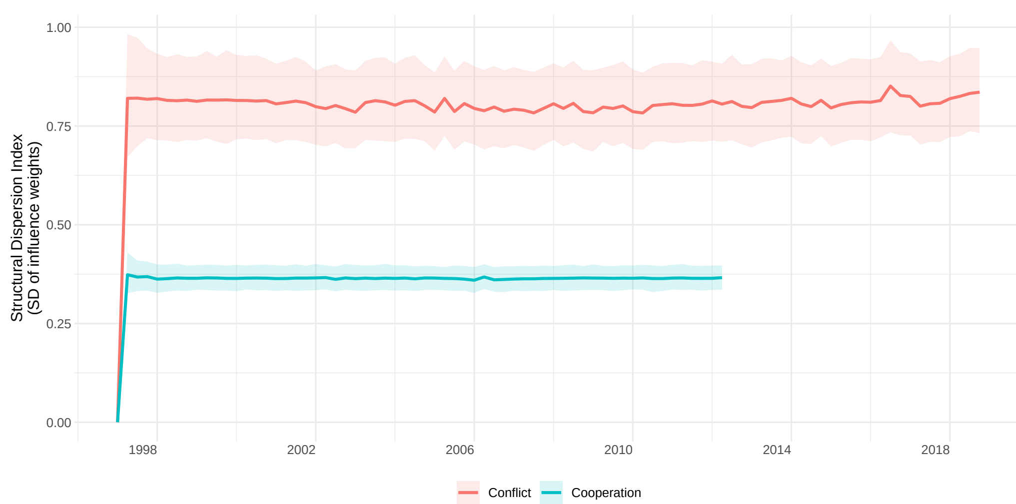

Figure A13: Structural dispersion index (standard deviation of off-diagonal influence weights $W_{ij,t}$) for Somalia's conflict and cooperation networks over time.

A.11 Cross-Case Comparison

We estimate dynamic bilinear models for the Democratic Republic of Congo (25 actors, 94 conflict quarters, 72 cooperation quarters) and Sudan (25 actors, 94 conflict quarters, 84 cooperation quarters) using the same sampler settings as for Somalia. Table A2 summarizes the network dimensions and estimated innovation variances across all six models.

Table A2: Cross-case summary of dynamic model estimates.

	n	T	$\hat{\tau}_A^2$
DRC Conflict	25	94	0.133
Somalia Conflict	25	88	0.169
Sudan Conflict	25	94	0.125
DRC Cooperation	25	72	0.117
Somalia Cooperation	34	62	0.085
Sudan Cooperation	25	84	0.117

Across all three countries, conflict networks exhibit larger innovation variances τ_A^2 than their cooperation counterparts, indicating greater temporal variation in the estimated influence mapping. Somalia's conflict network stands out with the highest innovation variance, consistent with its role as the primary illustrative case. This aligns with the Somalia coupling trajectories, which display a sustained structural reorganization rather than episodic spikes, whereas DRC and Sudan exhibit more punctuated shifts consistent with localized or phase-specific reweighting.

Figure A14 displays coupling trajectories for the two most-coupled actors in each conflict network. In all three cases, government forces are among the top two, and the leading non-state actors correspond to prominent armed organizations in the conflict histories: Al-Shabaab in Somalia, the FDLR in DRC, and the SPLM/A in Sudan.[22] The form

[22]The Sudan case carries the additional complication that the SPLM/A became the governing party

of temporal evolution varies across conflicts. Somalia exhibits the clearest sustained rise: both top actors' coupling norms are stable through the early years before diverging upward after approximately 2006, with government forces rising somewhat faster. DRC shows sharp coupling spikes for government forces during the Second Congo War period, with the FDLR maintaining a more stable but lower baseline. Sudan's government coupling remains moderate until a late spike, while the SPLM/A stays flat throughout. These differences are consistent with the variation in τ_A^2: Somalia, with the largest innovation variance, shows the most sustained structural change, while DRC and Sudan exhibit more episodic or localized shifts.

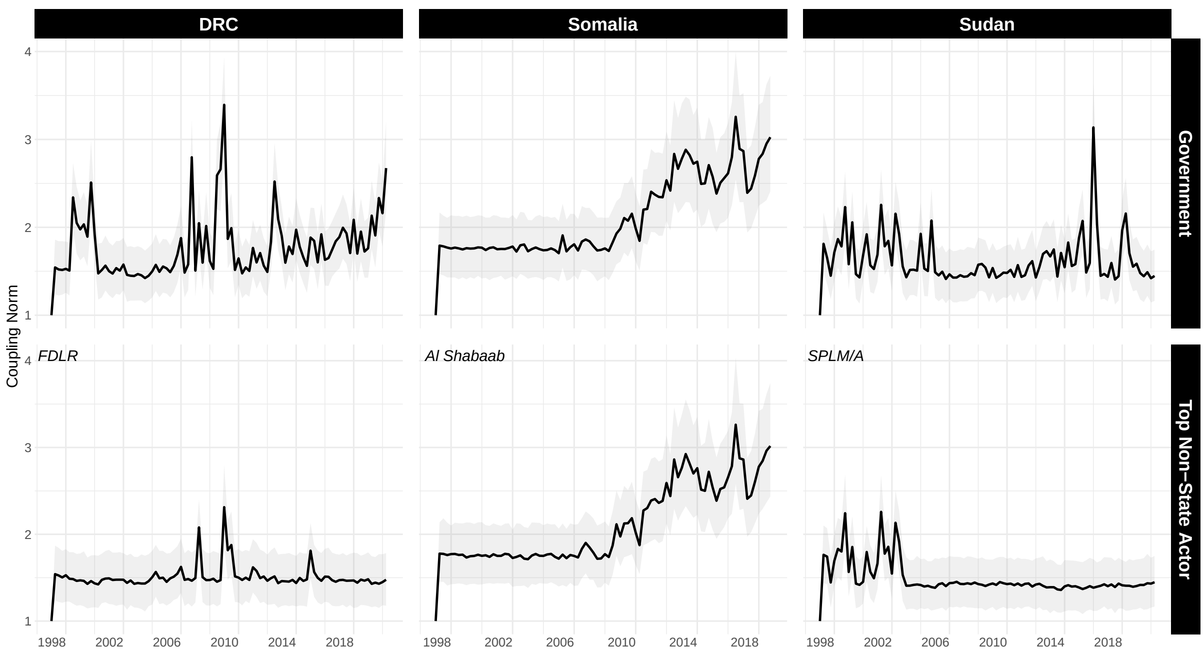

Figure A14: Coupling trajectories ($\|A_{t,i\cdot}\|_2$) for the two most-coupled actors in each conflict network. Top row: government forces; bottom row: top non-state actor (labeled in each panel). Lines show posterior medians; shaded regions show 80% credible intervals.

of independent South Sudan in 2011. The SPLM/A's high coupling in our estimates reflects primarily its pre-2011 position; the post-2011 period is better characterized as an interstate dynamic that our domestic-actor framing does not fully capture. The Sudan results should be interpreted with this structural break in mind.

A.12 Propagation Profile Robustness

We present a robustness check for the propagation profile analysis in the main text.

A.12.1 Varying the perturbation target

The main-text propagation profiles perturb the tie between each focal actor and the government. To verify that the results are not an artifact of this particular target choice, we repeat the analysis using each actor's modal opponent, the actor with whom they share the most conflict ties across the sample. For Al-Shabaab, the modal opponent is the government itself, so the two analyses coincide. For ASWJ, the modal opponent is Al-Shabaab; for Habar Gedir, it is the Biyamal Clan Militia.

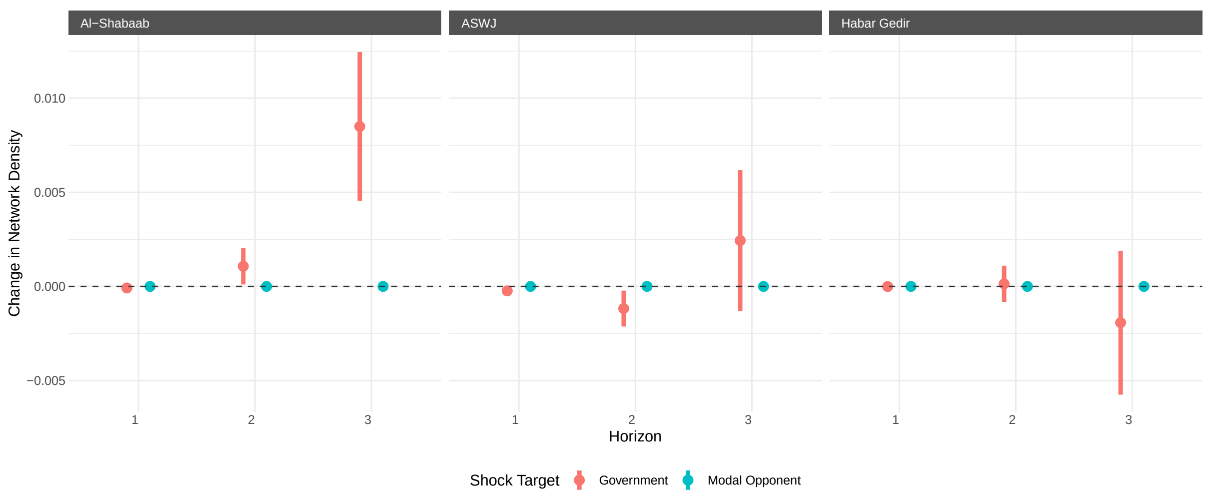

Figure A15: Propagation profile comparison: government target (red) vs. modal opponent target (blue). Points show posterior means; bars show posterior uncertainty.

Figure A15 reveals that government-targeted perturbations consistently produce larger and more precisely estimated downstream propagation than perturbations to peripheral opponents. This is substantively coherent: the government is the central node through which most conflict activity flows, so perturbations to government ties propagate through the network more broadly than perturbations to peripheral dyads.